

EVERY VALUATION RING IS F -SMOOTH

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Abstract

We establish two results uniformly in the prime p . For a henselian valuation ring V of mixed characteristic $(0, p)$ and for every $n \geq 0$, the mod- p syntomic-to-étale comparison

$$\mathbb{F}_p(n)_{\text{syn}}(V) \longrightarrow R\Gamma_{\text{ét}}(V[1/p], \mu_p^{\otimes n})$$

is an isomorphism in degrees $< n$ and injective in degree n . Moreover, every valuation ring is F -smooth at every prime, with no restrictions on rank, value group, or completeness.

The proof reduces to complete rank-one valuation rings. A Frobenius–Witt calculation yields weak F -smoothness; a Tor-unital fiber product, Dennis–Stein operations, and an all-degree normal-form argument establish the Milnor endpoint. After Frobenius base change, a geometric trivialization of the Breuil–Kisin line identifies twisted Nygaard-completion defects, which vanish by the endpoint theorem and Bott localization. The argument includes $p = 2$.

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1 Introduction

1.1 Main theorems

For a p -complete ring R , write

$$S_n(R) = \mathbb{F}_p(n)_{\text{syn}}(R), \quad E_n(R) = R\Gamma_{\text{ét}}(R[1/p], \mu_p^{\otimes n}),$$

and let $\gamma_{R,n} : S_n(R) \rightarrow E_n(R)$ be the natural comparison. For a noncomplete valuation ring, we use the standard extension of syntomic cohomology by the completion/generic-fiber pullback, as in [1].

Theorem 1.1 (Universal valuative comparison). *Let p be any prime and let V be a henselian valuation ring of mixed characteristic $(0, p)$. For every $n \geq 0$,*

$$\text{cofib}(S_n(V) \xrightarrow{\gamma_{V,n}} E_n(V)) \in D^{\geq n}(\mathbb{F}_p).$$

Equivalently, $\gamma_{V,n}$ is an isomorphism on H^q for $q < n$ and injective on H^n .

The theorem is uniform in the prime, rank, value group and weight. In particular, no discreteness, no noetherian hypothesis and no assumption $p > 2$ are imposed.

Theorem 1.2 (Universal valuation-ring F -smoothness). *Every valuation ring is F -smooth at every prime in the sense of Bhatt–Mathew. No condition on rank, value group or completeness is required.*

1.2 Proof architecture

The proof is organized so that each difficult point is isolated in a reusable statement. We give the arguments for the new derived, valuative and Milnor-theoretic steps; standard results from prismatic cohomology, syntomic cohomology and valuation theory are cited.

Step 1. Reduction and low-degree surjectivity. By valuative excision of Bachmann–Elmanto–Morrow, it suffices to treat complete rank-one valuation rings. A generic Milnor symbol of cohomological degree $q < n$ is lifted through a semistable regular local model, giving surjectivity of $H^q S_n \rightarrow H^q E_n$.

Step 2. Weak Frobenius smoothness. With $A = V/p$, the Nygaard–Sen triangle and Mao’s Frobenius–Witt cotangent complex compute the Frobenius pullback of

$$Q_i(V) = \text{cofib}(\theta : \text{gr}_N^i \Delta_V/p \rightarrow \text{gr}_N^{i+p} \Delta_V/p).$$

For nondiscrete rank one this is a direct sum of flat exterior powers in degrees $i+1, \dots, i+p$. Frobenius pullback reflects lower Tor bounds, so V is weakly F -smooth.

Step 3. Completion defects and first-overlap separation. Finite Bott strictness identifies the fiber of Bott multiplication on prismatic terms with a Hodge–Tate quotient. The twisted Nygaard filtration is shown separated on the first perfectoid Čech overlap. A uniform separation estimate for complete intersections then yields

$$E_k(V) := \text{cofib}(\bar{\Delta}_V\{k\}/p \rightarrow \widehat{\Delta}_V\{k\}/p) \in D^{\geq 1}$$

for every twist k .

Step 4. The Milnor endpoint. For a nondiscrete rank-one V with residue field k , the fiber product $T = V \times_k V$ is Tor-unital, and syntomic cohomology satisfies excision. A degree-three Dennis–Stein operation is extended to all Milnor degrees, and a graph-normal-form lemma gives

$$\widehat{K}_n^M(V)/p \hookrightarrow K_n^M(\text{Frac } V)/p$$

for every n , hence injectivity in the endpoint degree.

Step 5. Twist periodicity after Frobenius base change. Put $\psi : V \rightarrow A$, $v \mapsto \bar{v}^p$. On deriving pullback by ψ , the Breuil–Kisin line is trivialized via the Hodge–Tate stack and a two-dimensional regular prism

$$\mathbb{Z}_p[[T, B]]/(p - T^p B) \longrightarrow V.$$

The resulting unit identifies every pair ψ^*E_k and ψ^*E_ℓ .

Step 6. Vanishing of the completion defects. Starting from $E_k \in D^{\geq 1}$, the truncated Bott-defect comparison, Bott localization, low-degree surjectivity and the endpoint give one additional vanishing degree for one twist. Frobenius-pulled twist periodicity propagates it to all twists. Induction gives $E_k = 0$ for every k .

Step 7. Valuative comparison. When the defects vanish, Bott multiplication is injective on syntomic cohomology through the weight. Its telescope computes étale cohomology; low-degree surjectivity then finishes the complete rank-one case. Valuative excision transfers it to arbitrary henselian valuation rings.

Step 8. Universal F -smoothness. Section 10 extends the rank-one weak F -smoothness and actual Nygaard-completeness results to valuation rings of arbitrary rank, then removes the p -completeness hypothesis.

1.3 Conventions

We use prismatic and syntomic conventions from Bhatt–Lurie [2], Hodge–Tate and Nygaard calculations of Bhatt–Scholze [7], and the Bott-class and F -smoothness formalism of Bhatt–Mathew [4]. Unless flatness is established, tensor products, completions, fibers and cofibers are derived. Complexes are cohomologically graded, so $M[-r]$ lies in degree r for a discrete module M .

For a p -quasisyntomic ring R , Δ_R denotes absolute prismatic cohomology, $\mathcal{N}^{\geq r}\Delta_R$ its Nygaard filtration, and $\Delta_R\{k\}$ the k th Breuil–Kisin twist. A hat denotes Nygaard completion. Write $\bar{\Delta}_R\{k\} = \Delta_R\{k\}/I$ for Hodge–Tate specialization, with I the prism ideal. The universal degree- p Nygaard class is θ , and the mod- p syntomic Bott class is v_1 , of twist $p - 1$.

2 Reduction to complete rank one and surjectivity below the weight

2.1 The valuative reduction

The following reduction is the form of [1, Remark 8.4] used here; its proof rests on the v -excision square of [1, Proposition 7.25].

Proposition 2.1 (Bachmann–Elmanto–Morrow). *Fix p and n . If*

$$\mathrm{cofib}(S_n(W) \rightarrow E_n(W)) \in D^{\geq n}$$

for every p -complete henselian rank-one valuation ring W of mixed characteristic, the same holds for every henselian mixed-characteristic valuation ring.

Proof. We recall the mechanism needed later. Completion does not change the comparison cofiber, by the completion/generic-fiber definition of syntomic cohomology. For a complete valuation ring V , put $\mathfrak{q} = \sqrt{p}V$. The localization $V_{\mathfrak{q}}$ has rank one and the same p -inverted fraction field. The v -excision square compares V , $V_{\mathfrak{q}}$, $V/\mathfrak{q}$ and $\kappa(\mathfrak{q})$. The characteristic- p terms have comparison cofibers in $D^{\geq n}$; in degree n this uses injectivity of logarithmic forms from a Cartier-smooth valuation ring to its fraction field. Hence connectivity transfers from $V_{\mathfrak{q}}$ to V . See [1, Proposition 7.25 and Remark 8.4]. $\square$

Throughout Sections 2–9, V is a p -complete henselian rank-one valuation ring of mixed characteristic with fraction field K , unless specified otherwise.

2.2 A density lemma

Lemma 2.2. *Let $\Gamma \subset \mathbb{R}$ be a nondiscrete nonzero ordered subgroup. For every $m \geq 1$, every coset of $m\Gamma$ is dense in $\mathbb{R}$. Consequently, if K is a rank-one valued field with nondiscrete value group, every class of $K^{\times}/K^{\times m}$ has representatives of arbitrarily small positive valuation.*

Proof. A nondiscrete subgroup of $\mathbb{R}$ contains arbitrarily small positive elements and is therefore dense. Multiplication by m and translation preserve density. If z represents the desired class, choose $\gamma \in \Gamma$ with $v(z) + m\gamma$ in the required interval, then choose $h \in K^{\times}$ with $v(h) = \gamma$; zh^m has the required value. $\square$

2.3 The perfectoid quasisyntomic cover

The prismatic arguments use one canonical cover of a complete rank-one valuation ring.

Lemma 2.3 (Perfectoid valuation cover). *Let C be the completion of an algebraic closure of K , and put $W = \mathcal{O}_C$. The map $V \rightarrow W$ is a p -completely faithfully flat quasisyntomic cover. For every $a \geq 0$,*

$$A_a = (W^{\otimes_V(a+1)})_p^{\wedge}$$

is a discrete p -torsion-free quasiregular semiperfectoid ring. Absolute prismatic cohomology, its finite Nygaard quotients and Hodge–Tate specializations, and syntomic cohomology satisfy quasisyntomic descent on the resulting Čech nerve.

Proof. For an extension of valued fields, the cotangent complex of the extension of valuation rings has no homology above degree one, and its degree-one homology is torsion-free; in characteristic zero the stronger perfect-base form applies. This is the valuation cotangent theorem of Gabber–Ramero [8, Theorem 6.5.8]. Since torsion-free modules over a valuation domain are flat and every module over a valuation domain has flat dimension at most one, it gives the quasisyntomic Tor bound for $V \rightarrow W$. The valuation extension is faithfully flat, and p -adic completion preserves its reductions modulo p^r .

The tensor powers are flat over V , hence p -torsion-free; their derived p -completions are discrete and remain p -torsion-free. Quasisyntomicity is preserved by the relevant base changes. To see semiperfectoidness, choose compatible p -power roots in the algebraically closed perfectoid valuation ring W of elements in the additional tensor factors. A perfectoid root-polynomial algebra over the first copy of W surjects modulo p onto the tensor product; successive p -adic lifting gives a surjection onto its completion. Thus every A_a is QRSP.

Quasisyntomic descent follows from [7, Construction 7.6 and Theorem 12.2] and [2, Section 5.5]. Syntomic descent then follows from its description as the Frobenius-minus-canonical-map fiber [2, Proposition 7.4.6]. $\square$

2.4 Prime-to- p cyclotomic descent

We will repeatedly adjoin μ_p . The point is that this can be undone by a finite-stage trace even when the integral closure is not finite over V .

Lemma 2.4. *Let V be henselian of rank one, let L/K be cyclic of degree m prime to p , and suppose $\mu_m \subset K$. If V' is the valuation ring of L , there is a finite étale valuation extension U/V such that V' is a filtered union of finite-free U -algebras of the form $U[t]$, each with fraction field L . Their rank over V is m .*

Proof. Choose a Kummer generator r with $r^m \in K$. Let e be the order of $v(r)$ in Γ_L/Γ_K and put $f = m/e$. After multiplying r^e by a scalar in $K^\times$, obtain a unit s satisfying $s^f \in V^\times$. Henselianity and the prime-to- p hypothesis show that $U = V[s]$ is finite étale of rank f ; a valuation-basis argument identifies it with the valuation ring of $K(s)$.

Now $L = \text{Frac}(U)(r)$ has degree e , and the values of $1, r, \dots, r^{e-1}$ are distinct modulo Γ_K . In the nondiscrete case, replace r by br with $b \in K(s)^\times$ and $v(br) > 0$ arbitrarily small. Any prescribed finite subset of V' lies in the algebra $U[br]$, and these algebras form a directed system. In the discrete case choose a Kummer generator of minimal positive value and obtain $V' = U[r]$. $\square$

For a monic polynomial algebra $B = A[T]/(f)$ of degree d , syntomic Gysin maps and the projective bundle formula give a trace

$$\text{Tr}_f : S_i(B) \longrightarrow S_i(A)$$

whose composite with restriction is multiplication by d ; on a finite separable generic fiber it realizes étale corestriction. This follows from [16, Theorem 5.22] and [2, Theorem 9.3.1]. We only use the construction for the finite-free polynomial quotients of Lemma 2.4.

Proposition 2.5 (Cyclotomic descent). *Let $L = K(\mu_p)$ with valuation ring V' . For every i, q :*

- (1) $H^q S_i(V) \rightarrow H^q S_i(V')$ is injective;
- (2) an étale class over K whose restriction to L has a syntomic lift over V' has a syntomic lift over V ;
- (3) injectivity of $H^q S_i(V') \rightarrow H^q E_i(V')$ implies injectivity over V .

Proof. The degree $m = [L : K]$ divides $p - 1$. By Lemma 2.4, V' is a filtered union of finite-free polynomial quotients B_t of total rank m over V , after a finite étale intermediate extension. Mod- p syntomic cohomology commutes with filtered colimits [2, Corollary 8.4.11]. The traces give

compatible maps

$$T_t : H^q S_i(B_t) \longrightarrow H^q S_i(V), \quad T_t \circ \text{res} = m,$$

compatible with generic corestriction. Since $m \in \mathbf{F}_p^\times$, (1) follows. If α_t lifts the restriction of an étale class β , then

$$\gamma(T_t \alpha_t) = \text{Cor}_{L/K} \text{Res}_{L/K}(\beta) = m\beta,$$

which proves (2) after multiplying by m^{-1} . Part (3) follows from (1). For $p = 2$, $m = 1$. $\square$

2.5 Semistable lifting below the weight

Proposition 2.6 (Surjectivity below the weight). *For every $i \geq 1$ and $0 \leq q < i$, the comparison*

$$H^q S_i(V) \longrightarrow H_{\text{ét}}^q(K, \mu_p^{\otimes i})$$

is surjective.

Proof. By Proposition 2.5 we may assume $\mu_p \subset K$ and fix a primitive root ζ . If the valuation is discrete, V is a complete DVR and the assertion follows from the regular comparison theorem [4, Theorem 4.15 and Proposition 5.2]. Assume the value group is nondiscrete and put $\lambda = \zeta - 1$. By the norm-residue theorem [18, Theorem 6.16], the target is generated by classes

$$\zeta^{\otimes(i-q)} \cup [a_1] \cup \cdots \cup [a_q].$$

By Lemma 2.2, replace each a_r in its p th-power class so that

$$0 < v(a_r) < \frac{v(\lambda)}{q+1}.$$

Then $d = \lambda/(a_1 \cdots a_q)$ lies in the maximal ideal. Consider the henselian regular local ring

$$R = (\mathbf{Z}_p[\zeta][X_1, \dots, X_q, Y]/(X_1 \cdots X_q Y - \lambda))_{(X_1, \dots, X_q, Y)}^h.$$

It is regular of dimension $q+1$ and the assignments $X_r \mapsto a_r$, $Y \mapsto d$ define a local map $R \rightarrow V$. After inverting p , the X_r are units. Every regular noetherian ring is F -smooth, and on a p -torsion-free F -smooth ring the syntomic-to-étale comparison is an isomorphism below the weight [4, Theorem 4.15 and Proposition 5.2]. Since $q < i$, the generic symbol has a syntomic lift on R , whose pullback gives the desired lift on V . For $p = 2$, $\lambda = -2$. $\square$

3 Weak Frobenius smoothness of complete rank-one valuation rings

The aim of this section is to prove a weight-uniform lower bound for the Bott defect on Nygaard graded pieces. The proof is different in the discrete and nondiscrete cases; the latter is the essential one.

3.1 The universal Bott-defect triangle

For an animated ring R , put

$$A_R = R \otimes_{\mathbf{Z}}^{\mathbf{L}} \mathbf{F}_p, \quad C_R = L_{A_R/\mathbf{F}_p}, \quad K_n(R) = \bigwedge^n C_R[-n].$$

Write

$$G_n(R) = \mathrm{gr}_N^n \Delta_R/p, \quad Q_i(R) = \mathrm{cofib}(\theta : G_i(R) \rightarrow G_{i+p}(R)).$$

The Nygaard–Sen description of $\mathrm{gr}_N^n \Delta_R$ gives a fiber sequence

$$\mathrm{gr}_N^n \Delta_R \longrightarrow \mathrm{Fil}_n^{\mathrm{conj}} \widehat{\Omega}_R^D \xrightarrow{\Theta+n} \mathrm{Fil}_{n-1}^{\mathrm{conj}} \widehat{\Omega}_R^D,$$

where the n th conjugate graded piece is $(\bigwedge^n L_{R/\mathbf{Z}})_p^\wedge[-n]$ and Θ acts there by $-n$; see [2, Construction 4.7.1 and Remark 5.5.8]. We also use the polynomial-variable decompositions of [2, Example 5.6.10 and Variants 5.6.12–5.6.13].

Lemma 3.1 (Universal triangle). *For every animated ring R and $i \geq 0$ there is a natural fiber sequence in $D(A_R)$*

$$Q_i(R) \xrightarrow{q_i} K_{i+p}(R) \xrightarrow{\beta_i} K_i(R).$$

The construction is compatible with adjoining polynomial variables: under

$$K_n(R[X]) \simeq K_n(R)[X] \oplus K_{n-1}(R)[X][-1],$$

the map $\beta_i(R[X])$ decomposes as $\beta_i(R)[X] \oplus \beta_{i-1}(R)[X][-1]$.

Proof. The multiplicative Hodge projection gives $h_n : G_n(R) \rightarrow K_n(R)$. The Hodge image of θ is canonically zero, since it is pulled back from $\mathbf{Z}_p$ and $K_p(\mathbf{Z}_p) = 0$. Thus h_{i+p} descends to $q_i : Q_i \rightarrow K_{i+p}$. We retain q_i and its specified nullhomotopy before making cohomological calculations; this is needed under animation and for identifying the scalar on exterior-power summands.

Suppose first that R is a polynomial $\mathbf{Z}$ -algebra. Its conjugate filtration is the Postnikov filtration. Writing $\widehat{\Omega}_R^r = (\Omega_{R/\mathbf{Z}}^r)_p^\wedge$, the Sen sequence gives, for $n > 0$,

$$H^s(\mathrm{gr}_N^n \Delta_R) = \begin{cases} \widehat{\Omega}_R^{s-1}/(n-s+1), & 1 \leq s < n, \\ \widehat{\Omega}_R^n, & s = n, \\ 0, & s \leq 0 \text{ or } s > n. \end{cases}$$

Indeed, in degree $s-1 < n$, $\Theta+n$ acts by the nonzero integer $n-s+1$ on the p -torsion-free module $\widehat{\Omega}_R^{s-1}$; the top Hodge projection identifies the degree- n group.

Set $n = i+p$. After derived reduction modulo p , the only cohomology of G_{i+p} in degrees greater than i occurs in degrees $i+1$ and $i+p$:

$$H^{i+1}(G_{i+p}) = \Omega_R^i/p, \quad H^{i+p}(G_{i+p}) = \Omega_R^{i+p}/p.$$

Among $n-s+1$ with $i < s < n$, the unique multiple of p is p , at $s = i+1$; the extra Tor term produced by reducing the p -torsion belongs to degree i and does not occur in this range. The top group is the reduction of the p -torsion-free module $\widehat{\Omega}_R^{i+p}$.

Because $G_i \in D^{\leq i}$, it does not change cofiber cohomology above i . The ring R is regular, so [4, Theorem 4.15] implies $Q_i \in D^{\geq i+1}$. The map q_i identifies the degree- $(i+p)$ group with $K_{i+p}(R)$. Put $F_i(R) = \text{fib}(q_i : Q_i(R) \rightarrow K_{i+p}(R))$. It has a single cohomology group in degree $i+1$, canonically identified with Ω_R^i/p ; hence

$$F_i(R) \simeq K_i(R)[-1]. \quad (1)$$

The normalization is given by the Sen cokernel: $H^{i+1}F_i$ is the cokernel of multiplication by p on $\widehat{\Omega}_R^i$. No splitting of the remaining terms of the Sen sequence is chosen.

Now animate the diagram of q_i , its nullhomotopy, its functorial fiber F_i , and (1). Modulo p , G_n preserves sifted colimits [2, Remark 5.5.10], as does K_n by the construction of derived exterior powers. Finite fibers and cofibers preserve sifted colimits in a stable category. Left Kan extension from polynomial rings therefore gives the coherent triangle for every animated ring. Rotating it defines β_i , together with the specified maps and nullhomotopies.

For polynomial compatibility, the decompositions over $R[X]$ are induced by the unit and exterior multiplication with dX . Multiplicativity identifies θ on each summand, and its Hodge-image nullhomotopy is compatible with dX . Thus there is a canonical decomposition

$$\begin{aligned} (Q_i(R[X]) \xrightarrow{q_i} K_{i+p}(R[X])) \\ \simeq (Q_i(R) \xrightarrow{q_i} K_{i+p}(R))[X] \oplus (Q_{i-1}(R) \xrightarrow{q_{i-1}} K_{i+p-1}(R))[X][-1]. \end{aligned}$$

Taking fibers gives the corresponding decomposition of $F_i(R[X])$. On the dX summand the Sen operator changes from Θ to $\Theta - 1$, giving the cokernel of multiplication by p on $\widehat{\Omega}_R^{i-1}dX$ in degree $i+1$. Under (1), the map is exterior multiplication with dX , with coefficient one up to the fixed differential-ordering convention. After shifting the three fiber terms to degree zero, they lie in the heart of the derived module category; mapping spaces between such objects are discrete. The differential-form maps therefore lift uniquely to complexes with their commuting squares. Animation yields the stated decomposition of $\beta_i(R[X])$. $\square$

3.2 The Frobenius–Witt splitting

We use Mao’s Frobenius–Witt cotangent complex [13]. For a ring R , write FL_R for the pointed R/p -module of Frobenius–Witt differentials. There is a natural triangle

$$A_R \longrightarrow \text{FL}_R \longrightarrow L_{R/\mathbf{Z}} \otimes_{R,F}^{\mathbf{L}} A_R \longrightarrow A_R[1]. \quad (2)$$

Mao proves that FL_V is a flat discrete V/p -module for every valuation ring V [13, Theorem A.2], and that $\text{FL}_R = 0$ when the derived p -completion of R is perfectoid [13, Lemma A.1]. For a discrete ring the generators $w_R(x)$ obey

$$\begin{aligned} w_R(xy) &= x^p w_R(y) + y^p w_R(x), \\ w_R(x+y) &= w_R(x) + w_R(y) - P(x, y)w_R(p), \\ w_R(1) &= 0, \end{aligned}$$

where $P(x, y) = ((x+y)^p - x^p - y^p)/p$; see [13, Section 2 and Theorem 3.6].

Lemma 3.2. *Assume V is nondiscrete; put $A = V/p$ and $M = \text{FL}_V$. Then $w_V(p) = 0$ and*

$$F_A^* C_V \simeq A[1] \oplus M.$$

Consequently, for every $n \geq 0$,

$$F_A^* K_n(V) \simeq \bigoplus_{r=0}^n \bigwedge^r M[-r], \quad (3)$$

and every exterior power in this formula is flat.

Proof. Normalize $v(p) = 1$. Density gives $a \in V$ with $p^{-2} \leq v(a) < p^{-1}$ and $b = p/a^p$. Then a, b are in the maximal ideal and $a^{p^2} \in pV$. The Frobenius–Witt relations give

$$w_V(a^p) = pa^{p(p-1)}w_V(a) = 0, \quad w_V(p) = w_V(a^pb) = a^{p^2}w_V(b) = 0.$$

Thus the first map in (2) vanishes. As M is discrete, the zero-point splitting is unique, giving $F_A^* C_V \simeq A[1] \oplus M$. Derived exterior powers and the usual décalage formula give (3). Flatness follows by writing M as a filtered colimit of finite free modules. $\square$

3.3 The scalar calculation

After Frobenius pullback, β_i is a unit on each common exterior-power summand. Choose a, b as above and put

$$R_0 = (\mathbf{Z}_p[T, B]/(p - T^p B))_{(T, B)}, \quad A_0 = R_0/p.$$

Evaluation $T \mapsto a, B \mapsto b$ gives a local map $R_0 \rightarrow V$. The ring R_0 is regular local of dimension two and p -torsion-free.

Lemma 3.3. *For every $j \geq 0$, the Frobenius pullback of $\beta_j(R_0)$, after scalar extension $A_0 \rightarrow A$, acts on the canonical H^0 copy of A by a unit $u_j \in A^\times$.*

Proof. By regularity and [13, Theorem 5.2], FL_{R_0} is flat over A_0 . The Frobenius–Witt relation gives $w_{R_0}(p) = T^{p^2}w_{R_0}(B)$; this vanishes after base change to A because $a^{p^2} = 0$ in A . Thus the splitting of Lemma 3.2 yields a distinguished H^0 line, on which $F^*\beta_j$ acts by a scalar u_j .

Extend K to its completed algebraic closure, with valuation ring W . Then W is perfectoid and $A \rightarrow W/p$ is local. Mao’s perfectoid vanishing yields $\mathrm{FL}_W = 0$, while Bhatt–Mathew’s perfectoid Nygaard calculation gives $Q_j(W) = 0$ [4, Example 2.5]. Therefore $\beta_j(W)$ is an equivalence. Naturality shows that u_j becomes a unit in W/p ; since $A \rightarrow W/p$ reflects units, u_j is already a unit. $\square$

Lemma 3.4. *Under the splitting (3), the map*

$$H^r(F_A^* \beta_i(V)) : \bigwedge^r M \longrightarrow \bigwedge^r M, \quad 0 \leq r \leq i,$$

is multiplication by the unit $\pm u_{i-r}$.

Proof. For $x_1, \dots, x_r \in V$, evaluate $R_0[X_1, \dots, X_r] \rightarrow V, X_s \mapsto x_s$. The polynomial compatibility in Lemma 3.1 identifies the pure r -differential summand of β_i with $\beta_{i-r}(R_0)[-r]$. Under Frobenius–Witt splitting, its generator maps to $w_V(x_1) \wedge \dots \wedge w_V(x_r)$. Thus the map is $\pm u_{i-r}$ on these wedges, which generate $\bigwedge_V^r M$. The sign is the fixed Koszul sign and is a unit also for $p = 2$. $\square$

3.4 Frobenius pullback reflects Tor bounds

Lemma 3.5. *Let $A = V/p$ for a rank-one mixed-characteristic valuation ring. Derived Frobenius pullback F_A^* reflects Tor-amplitude intervals and one-sided Tor lower bounds.*

Proof. Let W be the valuation ring of the completed algebraic closure of K , and put $D = W/p$. The map $A \rightarrow D$ is faithfully flat. Frobenius on D factors as

$$D \longrightarrow D/(\varpi) \xrightarrow{\sim} D, \quad \varpi^p = p.$$

Base change along a nilpotent quotient reflects Tor bounds: when $J^N = 0$, each discrete module has a finite filtration with R/J -module subquotients. Hence F_D^* reflects Tor bounds. Frobenius naturality gives

$$(F_A^* E) \otimes_A^{\mathbf{L}} D \simeq F_D^*(E \otimes_A^{\mathbf{L}} D).$$

A bound on $F_A^* E$ gives the same bound on $E \otimes_A^{\mathbf{L}} D$, which faithful flatness reflects to E . $\square$

3.5 Weak F -smoothness

Theorem 3.6. *Let V be a p -complete rank-one valuation ring of mixed characteristic. Then V is p -quasisyntomic and weakly F -smooth in the sense of [4, Definition 4.2]. If V is nondiscrete, for every $i \geq 0$,*

$$F_A^* Q_i(V) \simeq \bigoplus_{r=i+1}^{i+p} \bigwedge_A^r M[-r], \quad A = V/p, \quad M = \mathrm{FL}_V. \quad (4)$$

In particular $Q_i(V)$ has A -Tor amplitude bounded below by $i+1$. Moreover, for every twist $n \in \mathbf{Z}$ and $m \geq 0$,

$$\mathrm{fib}(\theta : \mathrm{gr}_N^m \Delta_V \{n\}/p \longrightarrow \mathrm{gr}_N^{m+p} \Delta_V \{n+p-1\}/p) \in D^{\geq m+2}, \quad (5)$$

$$\mathrm{fib}(\varphi_m : \mathrm{gr}_N^m \Delta_V \{n\}/p \longrightarrow \overline{\Delta}_V \{m\}/p) \in D^{\geq m+2}. \quad (6)$$

Proof. First suppose V is nondiscrete. Apply F_A^* to Lemma 3.1. Lemma 3.2 splits $F_A^* K_{i+p}(V) = L \oplus H$, where

$$L = \bigoplus_{r=0}^i \bigwedge_A^r M[-r], \quad H = \bigoplus_{r=i+1}^{i+p} \bigwedge_A^r M[-r].$$

By Lemma 3.4, the restriction of $F_A^* \beta_i$ to L is an equivalence. A change of coordinates removes the off-diagonal map from H to the target, so the fiber is H . This proves (4). All exterior powers of M are flat, so the lower Tor bound is $i+1$; Lemma 3.5 reflects it to $Q_i(V)$.

The same argument applied to $F_A^* C_V = A[1] \oplus M$ shows that $L_{A/\mathbf{F}_p}$ has Tor amplitude $[-1, 0]$. Since V is p -complete and p -torsion-free, this is the cotangent condition for p -quasisyntomicity. The fiber of $\theta : G_i \rightarrow G_{i+p}$ is $Q_i[-1]$. Changing rings from A to V loses at most one Tor degree, since p is a nonzerodivisor, giving the weak F -smoothness bound of [4, Definition 4.2]. In particular $Q_i(V) \in D^{\geq i+1}$, so the untwisted θ -fiber lies in $D^{\geq i+2}$.

For twists, the associated graded Breuil–Kisin line has a canonical degree-zero unit, and multiplication identifies each twisted graded piece with the untwisted one; see [2, Corollary 5.5.14 and Remark 5.5.15]. The graded image of the syntomic Bott lift is θ times this unit [4, Construction 2.8 and Proposition 2.9]. This gives (5). Divided Frobenius is the telescope obtained by inverting θ

on the Nygaard graded algebra [4, Proposition 2.6]. Successive cofibers have lower bounds tending to infinity, so the fiber from the first term to the telescope lies in $D^{\geq m+2}$, giving (6).

If V is discrete, it is a complete DVR and hence regular noetherian. The same assertions follow from the F -smoothness of regular rings [4, Theorem 4.15], along with the same graded-twist and telescope arguments. $\square$

4 Bott strictness and Nygaard-completion defects

We isolate the discrepancy between Bott multiplication on the uncompleted and completed prismatic terms. It is measured by the twist-indexed defects E_k and by their truncated comparison with Bott cofibers in syntomic cohomology.

4.1 Finite Bott strictness on QRSP rings

Let R be a p -torsion-free quasiregular semiperfectoid (QRSP) ring. Put

$$D = \Delta_R/p, \quad N^r = \mathcal{N}^{\geq r} \Delta_R/p, \quad L = \Delta_R\{1\}/p.$$

Locally choose a generator d of the prism ideal and a generator e of the Breuil–Kisin line, normalized by $\varphi_1(e) = (u/d)e$ with $u \in D^\times$. In this trivialization the mod- p Bott class is

$$b = ce^{p-1}, \quad c = d/u,$$

up to a nonzero scalar in $\mathbf{F}_p$; see [4, Constructions 2.7–2.8, Proposition 2.9 and Remark 2.10].

Lemma 4.1 (Finite Bott strictness). *Multiplication by c is injective on D and, for every $r \geq 0$,*

$$cD \cap N^{r+p} = cN^r. \tag{7}$$

Consequently, for every $q \geq p$ there is a short exact sequence

$$0 \longrightarrow D/N^{q-p} \xrightarrow{c} D/N^q \longrightarrow D/(cD + N^q) \longrightarrow 0. \tag{8}$$

These statements are intrinsic after restoring Breuil–Kisin twists.

Proof. For a p -torsion-free QRSP ring, divided Frobenius identifies the Nygaard graded pieces with the relative conjugate filtration; these pieces inject into D after reduction modulo p , by [7, Theorem 12.2 and Proposition 12.8]. Moreover (p, d) is a regular sequence on the prismatic ring, so c is a nonzerodivisor modulo p . Since

$$\frac{\varphi(c)}{d^p} = u^{-p} \in D^\times,$$

we have $c \in N^p$, and multiplicativity of divided Frobenius shows that $c : \mathrm{gr}_N^r D \rightarrow \mathrm{gr}_N^{r+p} D$ is injective. If $cy \in N^{r+p}$, begin with $y \in N^0 = D$ and induct on $s < r$. If $y \in N^s$, the image of cy in $\mathrm{gr}_N^{s+p} D$ vanishes, hence injectivity forces $y \in N^{s+1}$. Thus $y \in N^r$, proving (7). The quotient exact sequence follows. $\square$

4.2 The completion defect

For a twist k , put

$$H_k(R) = \overline{\Delta}_R\{k\}/p.$$

For $q \geq p$, let $B_q^k(R)$ be the intrinsic finite quotient, defined as the cofiber of Bott multiplication

$$\frac{\Delta_R\{k-p+1\}/p}{\mathcal{N}^{\geq q-p}} \xrightarrow{v_1} \frac{\Delta_R\{k\}/p}{\mathcal{N}^{\geq q}}.$$

On a QRSP chart Lemma 4.1 identifies this with the ordinary quotient $D\{k\}/(ID\{k\} + N^q D\{k\})$. For a general quasisyntomic ring this finite cofiber is the definition, compatible with quasisyntomic descent and filtered colimits modulo p . Define the induced filtration by

$$F^q H^0(H_k) = \ker(H^0 H_k \rightarrow H^0 B_q^k),$$

and set

$$\widehat{H}_k = \varprojlim_q B_q^k, \quad E_k(R) = \operatorname{cofib}(H_k(R) \longrightarrow \widehat{H}_k(R)).$$

Lemma 4.2 (Bott fiber versus completion defect). *Let $j \in \mathbf{Z}$ and $k = j + p - 1$. Define*

$$\begin{aligned} Y_j &= \operatorname{fib}(b : \Delta_R\{j\}/p \rightarrow \Delta_R\{k\}/p), \\ \widehat{Y}_j &= \operatorname{fib}(b : \widehat{\Delta}_R\{j\}/p \rightarrow \widehat{\Delta}_R\{k\}/p). \end{aligned}$$

Then $Y_j \simeq H_k[-1]$ and $\widehat{Y}_j \simeq \widehat{H}_k[-1]$, whence

$$\operatorname{fib}(Y_j \rightarrow \widehat{Y}_j) \simeq E_k[-2]. \quad (9)$$

For p -torsion-free QRSP R , the objects H_k and $\widehat{H}_k$ are discrete, and E_k has amplitude in $[-1, 0]$.

Proof. On a QRSP chart, multiplication by b is multiplication by the nonzerodivisor c after trivializing the twist. Its cokernel is the Hodge–Tate quotient H_k , giving the first equivalence. Taking limits in (8) is exact since all transition maps in the quotient towers are surjective, and gives the second. Reduction modulo p commutes with these derived limits since $\mathbf{Z}/p$ is perfect over $\mathbf{Z}$. Quasisyntomic descent globalizes the identifications. The fiber of the map between the Bott fibers is (9). $\square$

4.3 From graded bounds to completed Bott fibers

The weak F -smoothness bounds of Theorem 3.6 remain valid after Nygaard completion.

Lemma 4.3. *For $m \geq 0$ and every twist n ,*

$$\operatorname{fib}(v_1 : \mathcal{N}^{\geq m} \widehat{\Delta}_V\{n\}/p \rightarrow \mathcal{N}^{\geq m+p} \widehat{\Delta}_V\{n+p-1\}/p) \in D^{\geq m+2}. \quad (10)$$

Moreover divided Frobenius factors naturally as

$$\mathcal{N}^{\geq n} \widehat{\Delta}_V\{n\}/p \xrightarrow{\widetilde{\varphi}_n} H_n(V) \longrightarrow \widehat{H}_n(V),$$

with

$$\operatorname{fib}(\widetilde{\varphi}_n) \in D^{\geq n+2} \quad (n \geq 0). \quad (11)$$

Proof. The completed Nygaard filtration is complete by construction. The graded pieces of the fiber in (10) are the twisted θ fibers of (5), with indices at least m . Finite extensions and inverse limits preserve the common bound $m + 2$, proving (10). For divided Frobenius, the associated graded map is the map to the θ -localized Hodge–Tate complex. Equation (6) gives the lower bound on every graded fiber. The filtered factorization through the completed source is that of [2, Remarks 5.7.8 and 5.7.10]. Passing to complete filtered objects gives (11). $\square$

Put

$$C_j(V) = \text{cofib}(v_1 : S_j(V) \rightarrow S_{j+p-1}(V)).$$

The following truncated comparison is used both for the initial defect bound and in the induction of Section 9.

Proposition 4.4 (Truncated Bott–defect comparison). *Let $j \geq 1$ and $k = j + p - 1$. Then*

$$\tau_{\leq j-1} C_j(V) \simeq \tau_{\leq j-1} (E_k(V)[-1]). \quad (12)$$

Equivalently, for $1 \leq q \leq j$, $H^{q-1} C_j(V) \simeq H^{q-2} E_k(V)$.

Proof. Syntomic cohomology is computed on Nygaard-completed terms by

$$S_i(V) \simeq \text{fib}(\widehat{\varphi}_i - \text{can} : \mathcal{N}^{\geq i} \widehat{\Delta}_V\{i\}/p \rightarrow \widehat{\Delta}_V\{i\}/p)$$

by [2, Proposition 7.4.6]. Taking Bott fibers between the two syntomic squares gives

$$F_j := \text{fib}(S_j \rightarrow S_k) = \text{fib}(\alpha - \beta : X_j \rightarrow \widehat{Y}_j),$$

where X_j is the Bott fiber on completed Nygaard sources. The divided-Frobenius contribution factors as $X_j \xrightarrow{f} Y_j \xrightarrow{g} \widehat{Y}_j$, $\alpha = gf$, and (11) implies $\text{fib}(f) \in D^{\geq j+2}$. The canonical-map contribution β factors through

$$Z_j = \text{fib}(v_1 : \mathcal{N}^{\geq j-1} \widehat{\Delta}_V\{j\}/p \rightarrow \mathcal{N}^{\geq j+p-1} \widehat{\Delta}_V\{k\}/p),$$

which belongs to $D^{\geq j+1}$ by (10). Applying $\tau_{\leq j}$, the map f becomes an equivalence, and the factorization through Z_j makes β zero in $D^{\leq j}$. Therefore $\tau_{\leq j} F_j \simeq \tau_{\leq j} \text{fib}(g)$. By (9), $\text{fib}(g) = E_k[-2]$; since $C_j = F_j[1]$, shifting gives (12). $\square$

5 Separation on the first perfectoid overlap

We aim to prove the initial bound $E_k(V) \in D^{\geq 1}$. It suffices to establish separation of the induced Nygaard filtration on the first nontrivial term of the perfectoid Čech nerve. We make the estimate uniform in the conjugate degree when the number of complete-intersection equations is fixed.

5.1 A universal separation estimate

Let $W = \mathcal{O}_C$ for an algebraically closed perfectoid field of mixed characteristic, and consider

$$S_r = W\langle X_1^{1/p^\infty}, \dots, X_r^{1/p^\infty} \rangle / (X_1, \dots, X_r).$$

Give $\overline{\Delta}_{S_r}/p$ the relative conjugate filtration over W ; the induced Nygaard filtration is the image of the prismatic Nygaard steps.

Lemma 5.1 (Universal bound). *For every $s \geq 0$,*

$$\mathcal{N}^{\geq n}(\bar{\Delta}_{S_r}/p) \cap \mathrm{Fil}_s^{\mathrm{conj}}(\bar{\Delta}_{S_r}/p) = 0 \quad \text{if } n \geq ps + r(p-1) + p. \quad (13)$$

In particular, the induced Nygaard filtration is separated.

Proof. The QRSP calculation of [7, Lemmas 12.4 and 12.7] gives a basis indexed by $\mathbb{N}[1/p]^r$. In one variable the basis element e_u has Nygaard coefficient $t^{\max(n-\lfloor u \rfloor, 0)}$ and relative conjugate degree $\lfloor u/p \rfloor$. Tensoring gives the multivariable basis. If the total conjugate degree is at most s , then

$$\sum_j \lfloor u_j \rfloor \leq p \sum_j \lfloor u_j/p \rfloor + r(p-1) \leq ps + r(p-1).$$

The displayed bound on n forces the Nygaard coefficient to be divisible by t^p . On Hodge–Tate specialization modulo p one has $t^p = 0$. This proves the intersection statement; exhaustiveness of the conjugate filtration gives separatedness. $\square$

Lemma 5.2 (Uniform complete-intersection separation). *Let P be a p -torsion-free perfectoid W -algebra and $f_1, \dots, f_r \in P$ a sequence whose images are Koszul regular on P/p . Put $S = (P/(f_1, \dots, f_r))_p^\wedge$ and assume S is p -torsion-free. Then the image of $H^0(\mathcal{N}^{\geq n} \Delta_S \otimes^{\mathbf{L}} \mathbf{F}_p)$ in $\bar{\Delta}_S/p$ has zero intersection with $\mathrm{Fil}_s^{\mathrm{conj}}$ whenever $n \geq ps + r(p-1) + p$. The same conclusion holds for filtered colimits of such presentations with a uniform bound on r .*

Proof. First assume compatible p -power roots of the f_i have been chosen in P . Put

$$P_0 = W\langle X_1^{1/p^\infty}, \dots, X_r^{1/p^\infty} \rangle, \quad S_0 = P_0/(X_1, \dots, X_r),$$

and map $P_0 \rightarrow P$ by sending X_i^{1/p^m} to the chosen roots of f_i . Koszul regularity gives a discrete derived base change

$$S_0/p \otimes_{P_0/p}^{\mathbf{L}} P/p \simeq S/p. \quad (14)$$

Use the perfectoid base-change statement in the proof of [7, Proposition 12.8]: prismatic cohomology and the relative conjugate filtration commute with derived p -complete base change along perfectoid algebras, and the transported good filtration is Nygaard. Thus

$$\mathcal{N}^{\geq n} \Delta_S \simeq \mathcal{N}^{\geq n} \Delta_{S_0} \hat{\otimes}_{\Delta_{S_0}}^{\mathbf{L}} \Delta_S. \quad (15)$$

This is an assertion about filtered complexes, not merely their associated graded pieces.

For the universal factorization, write every multi-index uniquely as $u = a + pk$, with $0 \leq a_i < p$ and $k \in \mathbb{N}^r$; put $g_k = e_{pk}$. Multiplication by the finitely many e_a identifies the Hodge–Tate specialization as

$$\bar{\Delta}_{S_0}/p = \bigoplus_{k \in \mathbb{N}^r} (S_0/p)g_k, \quad \mathrm{Fil}_s^{\mathrm{conj}} = \bigoplus_{|k| \leq s} (S_0/p)g_k. \quad (16)$$

The coefficient in $e_a g_k = u_{a,k} e_{a+pk}$ is a unit: all new q -factorial factors have indices strictly between consecutive multiples of p . Thus (16) is a genuine S_0/p -module direct sum, not only a splitting of the associated graded objects. Write $H_{>s}$ for the complementary sum.

Reducing a Nygaard step of a general QRSP ring modulo the prism ideal may create negative cohomology, so one cannot transport the assertion using only H^0 at the universal source. Here [7, Lemma 12.7] gives the explicit universal formula

$$\mathcal{N}^{\geq n} \Delta_{S_0} = \bigoplus_u t^{\max(n - \sum_i \lfloor u_i \rfloor, 0)} A_{\mathrm{inf}}(W) e_u,$$

where $t = \varphi^{-1}(\xi)$ and ξ generates the prism ideal of W . Each coefficient ideal is abstractly free of rank one, since t is a nonzerodivisor. As (p, ξ) is a regular sequence on $A_{\text{inf}}(W)$, the Nygaard step is (p, ξ) -completely flat. Consequently

$$D_n := \mathcal{N}^{\geq n} \Delta_{S_0} \otimes_{A_{\text{inf}}(W)}^{\mathbf{L}} A_{\text{inf}}(W)/(p, \xi) \quad (17)$$

is discrete. The numerical estimate in Lemma 5.1 shows that the entire morphism of discrete P_0/p -modules $D_n \rightarrow \bar{\Delta}_{S_0}/p$ factors through $H_{>s}$ whenever $n \geq ps + r(p-1) + p$. Tensor over P_0/p with P/p . By (14), the low and high sums in (16) remain discrete complementary free S/p -modules after derived base change. Equations (15) and (17) identify the transported source with the Nygaard map for S after Hodge–Tate reduction. Although its source can now have negative cohomology, the whole morphism still factors through the discrete high summand, so its image on H^0 cannot meet the low summand.

Remove the roots by André’s flatness lemma [7, Theorem 7.14]. After a p -completely faithfully flat perfectoid extension $P \rightarrow P'$, all f_i acquire compatible roots. Regularity modulo p is preserved by this flat base change, and $S' = (P'/(f_1, \dots, f_r))_p^\wedge$ is a p -completely faithfully flat quasisyntomic extension of S . The conjugate filtration and its Hodge–Tate graded pieces base change along $S/p \rightarrow S'/p$, so $H^0(\bar{\Delta}_S/p) \rightarrow H^0(\bar{\Delta}_{S'}/p)$ is injective. Naturality gives the desired intersection statement for S .

Finally, for a filtered colimit of these complete intersections, finite Nygaard quotients and finite conjugate pieces commute with filtered colimits; both are built from finitely many Nygaard graded pieces, identified with finite conjugate pieces by divided Frobenius. A class and its witness of membership in a fixed Nygaard level occur at one finite stage. The uniform estimate there proves the filtered-colimit assertion. $\square$

5.2 The first overlap

Let C be the completion of an algebraic closure of K , $W = \mathcal{O}_C$, and $R = (W \otimes_V W)_p^\wedge$. We prove separation of the induced Nygaard filtration on $\bar{\Delta}_R/p$. We use Gabber–Ramero’s description of tame and degree- p extensions [8, Proposition 6.2.12, Corollary 6.2.14 and Proposition 6.3.11]. After passage to the maximal tame extension, each finite Galois extension is built from a tower of degree- p extensions. At each stage the valuation ring is the filtered union of finite-free algebras given by one monic polynomial equation.

Lemma 5.3. *Let V^t be the valuation ring in the maximal tame extension K^t/K . Then*

$$B = (W \otimes_V V^t)_p^\wedge$$

is p -torsion-free perfectoid.

Proof. The tensor product is flat over W , hence p -torsion-free. The relative Frobenius

$$(V^t/p) \otimes_{V/p, F}^{\mathbf{L}} V/p \longrightarrow V^t/p$$

is an isomorphism. After strict henselization a finite tame cyclic extension of prime degree $\ell \neq p$ has a basis with distinct value cosets. Every nontrivial coset lattice is a directed union of principal lattices with values approaching zero; Frobenius carries this system isomorphically to that of the p -multiple coset. The filtered union is therefore relatively perfect. Base change to W/p identifies Frobenius $B/\pi \rightarrow B/p$, where $\pi = p^{1/p} \in W$. Since π is a nonzerodivisor, [5, Lemma 3.10] implies that B is perfectoid. $\square$

Lemma 5.4 (Triangular finite models). *Let $K_0 \subset K_1 \subset \cdots \subset K_r$ be a tower of henselian rank-one valued fields; suppose each K_i is tamely closed and $[K_i : K_{i-1}] = p$. Write $A_i = K_i^+$. Then A_r is a filtered union of A_0 -subalgebras with triangular presentations*

$$A_0[X_1, \dots, X_r]/(g_1, \dots, g_r), \quad (18)$$

where $g_i \in A_0[X_1, \dots, X_i]$ is monic of degree p in X_i . Each is finite free of rank p^r over A_0 . After a p -torsion-free perfectoid base change $A_0 \rightarrow B$, the completed base changes are complete intersections with the same r equations, regular modulo p .

Proof. For a degree- p extension, the assertion is the filtered single-generator description of [8, Proposition 6.3.11]: when $K = K^t$ and $[E : K] = p$, the valuation ring E^+ is a filtered union of finite K^+ -algebras $K^+[x]$ with integral primitive generator x and minimal polynomial monic of degree p .

Proceed inductively. For a finite subset $F \subset A_r$, choose an integral primitive element x_r for K_r/K_{r-1} such that $F \subset A_{r-1}[x_r]$. Only finitely many coefficients occur in the minimal polynomial and in the expressions of elements of F . By induction they lie in a single triangular algebra $D_{r-1} = A_0[x_1, \dots, x_{r-1}] \subset A_{r-1}$. Its fraction field is K_{r-1} . The minimal polynomial $g_r \in D_{r-1}[X_r]$ is monic of degree p , and the map

$$D_r := D_{r-1}[X_r]/(g_r) \longrightarrow A_r$$

is injective: D_r is free of rank p over D_{r-1} , and its generic fiber is the field K_r . It contains F . Every finite subset thus lies in a model (18), and two such models embed in a third by applying the construction to the union of their generators. Successive monic bases $1, X_i, \dots, X_i^{p-1}$ show that every model is finite free of rank p^r over A_0 .

Choose polynomial representatives for coefficients of g_i in the preceding variables. Monicity makes g_i a nonzerodivisor after quotienting by earlier equations and after arbitrary base change. Hence $(g_1, \dots, g_r)$ is a regular sequence, also modulo p when the resulting algebra is p -torsion-free. The completed perfectoid base changes remain finite free, and provide complete intersections with exactly r equations.

In the application, intermediate fields remain tamely closed: for separable E/K , $E^t = EK^t$ [8, Section 6.2.18]; if $K = K^t$ and E/K occurs in a finite p -tower, this identity gives $E = E^t$. $\square$

Proposition 5.5 (First-overlap separation). *The induced Nygaard filtration on $H^0(\overline{\Delta}_R/p)$ is separated.*

Proof. Let L/K range through finite subextensions of the algebraic closure and put $T_L = (W \otimes_V V_L)^\wedge_p$. Every Hodge–Tate class on R comes from one T_L , because $R/p = \text{colim}_L T_L/p$ and the finite conjugate filtration commutes with this colimit.

After replacing L by a finite Galois extension, set $E = LK^t$. The extension E/K^t is a finite p -extension. Choose a tower

$$K^t = F_0 \subset F_1 \subset \cdots \subset F_r = E, \quad [F_i : F_{i-1}] = p.$$

The intermediate fields remain tamely closed. Lemma 5.4 supplies a cofinal filtered system of subalgebras of E^+ with presentations $V^t[X_1, \dots, X_r]/(g_1, \dots, g_r)$, where g_i is monic of degree p in its new variable. After base change along $V^t \rightarrow B$ and p -completion these are p -torsion-free

complete intersections, regular modulo p , with precisely r equations. In particular this same r works for the filtered system attached to the fixed finite extension.

The map

$$T_L \longrightarrow (B \otimes_{V^t} E^+)_p^\wedge$$

is p -completely faithfully flat and quasisyntomic: valuation extensions are flat, and the cotangent complex has the Tor bound of [8, Theorem 6.5.8]. The target is a filtered colimit of these complete intersections. By Lemma 5.2, its Nygaard filtration is separated with a bound depending on the fixed tower length r and the conjugate degree. Quasisyntomic descent for finite quotients reflects separation to T_L .

To descend without changing the finite stage, let $T \rightarrow U$ be a p -completely faithfully flat quasisyntomic map. Both $\bar{\Delta}_T/p$ and every finite quotient $B_q(T)$ are connective, by the Hodge–Tate filtration and the quasisyntomic Tor bound. Čech descent makes both maps $H^0(\bar{\Delta}_T/p) \rightarrow H^0(\bar{\Delta}_U/p)$ and $H^0 B_q(T) \rightarrow H^0 B_q(U)$ injective. Thus the induced filtration is strict on H^0 :

$$H^0(\bar{\Delta}_T/p) \cap F^q H^0(\bar{\Delta}_U/p) = F^q H^0(\bar{\Delta}_T/p). \quad (19)$$

Let $x \in H^0(\bar{\Delta}_R/p)$ lie in every Nygaard level; choose L so that it comes from $x_L \in H^0(\bar{\Delta}_{T_L}/p)$. The map $T_L \rightarrow R$ is a p -completely faithfully flat quasisyntomic valuation base change. By (19), x_L lies in every Nygaard level of T_L . Separation there gives $x_L = 0$ and hence $x = 0$. $\square$

5.3 The initial defect bound

Let W be the perfectoid valuation cover of V and $A_a = (W^{\otimes_V(a+1)})_p^\wedge$ its Čech nerve. By Lemma 2.3 these rings are p -torsion-free QRSP and the relevant prismatic and syntomic functors satisfy descent.

Proposition 5.6 (Initial connectivity of the defects). *If V is nondiscrete, then $E_k(V) \in D^{\geq 1}$ for every $k \in \mathbf{Z}$.*

Proof. On each QRSP term Lemma 4.2 gives $E_k(A_a) \in D^{[-1,0]}$. The ring $A_0 = W$ is perfectoid, so its induced filtration is complete and $E_k(A_0) = 0$. By Proposition 5.5, the induced filtration on A_1 is separated, hence $H^{-1}E_k(A_1) = 0$. Descent gives $E_k(V) \simeq \text{Tot}(E_k(A_\bullet))$ and the spectral sequence

$$E_1^{a,b} = H^b E_k(A_a) \implies H^{a+b} E_k(V).$$

Only rows $b = -1, 0$ occur. In total degree zero, the only possibilities $(0, 0)$ and $(1, -1)$ vanish by the observations above; in total degree -1 only $(0, -1)$ can occur and it also vanishes. Thus $E_k(V) \in D^{\geq 1}$. $\square$

6 The Milnor endpoint and the fiber product

The other independent input to the final induction is endpoint injectivity. After the syntomic excision statement below, the argument is algebraic. We first construct the degree-two and three operations that will extend to all degrees.

6.1 Milnor and Dennis–Stein conventions

For a local ring R , let $K_*^M(R)$ be ordinary Milnor K -theory and $\widehat{K}_*^M(R)$ Kerz’s improved theory. For each fixed m there exists M_m , independent of R , such that

$$|R/\mathfrak{m}_R| > M_m \implies K_m^M(R) \xrightarrow{\sim} \widehat{K}_m^M(R); \quad (20)$$

see [11, Proposition 10(5)]. We enlarge the residue field only after fixing a finite range of degrees.

We use the absolute Dennis–Stein presentation of K_2 . For a 3-fold stable commutative ring, $K_2(R)$ is generated by symbols $\langle a, b \rangle$, defined when $1 + ab \in R^\times$, with relations

$$\langle a, b \rangle = -\langle -b, -a \rangle, \quad (21)$$

$$\langle a, b \rangle + \langle a, c \rangle = \langle a, b + c + abc \rangle, \quad (22)$$

$$\langle a, bc \rangle = \langle ab, c \rangle + \langle ac, b \rangle. \quad (23)$$

This is van der Kallen’s theorem [17, Theorem 7.1]. If R is local with more than five residue elements, it is 5-fold stable and the symbol map is an isomorphism

$$K_2^M(R) \xrightarrow{\sim} K_2(R), \quad (24)$$

by [17, Theorem 8.4 and Corollary 8.5]. In particular, for units u, v ,

$$\{u, v\} = -\{v, u\}, \quad \{u, -u\} = 0, \quad \{u, u\} = \{u, -1\}. \quad (25)$$

These identities are integral. In particular the last remains valid modulo 2, and the class $\{-1\}$ must then be retained.

For this section let V be nondiscrete, p -complete and rank one, with maximal ideal $\mathfrak{m}$, residue field k , and fraction field K . After fixing the degree range, first assume k is large enough for the relevant comparisons. Set

$$\begin{aligned} G &= K^\times / K^{\times p}, & P &= \ker(A_1 \rightarrow k^\times / k^{\times p}), \\ A_m &= K_m^M(V)/p, & A &= \bigoplus_{m \geq 0} A_m, & J &= PA \subset A, \end{aligned} \quad (26)$$

and let $e = \{-1\} \in A_1$. Every element of P is represented by a principal unit. By Lemma 2.2, each $g \in G$ has representatives of arbitrarily small positive valuation.

6.2 The fiber product and excision

Put $T = V \times_k V$ with projections $q_0, q_1 : T \rightarrow V$ and diagonal section $s : V \rightarrow T$.

Lemma 6.1 (Tor-unitality of the fiber product). *The square*

$$\begin{array}{ccc} T & \xrightarrow{q_0} & V \\ q_1 \downarrow & & \downarrow \\ V & \longrightarrow & k \end{array}$$

is a Tor-unital Milnor square. Consequently the square of nonconnective algebraic K -theory spectra is homotopy cartesian, and

$$\ker(K_2(T) \xrightarrow{q_0} K_2(V)) \simeq K_2(V, \mathfrak{m}), \quad (27)$$

compatibly with q_1 .

Proof. Nondiscreteness implies $\mathfrak{m}^2 = \mathfrak{m}$: for $0 \neq a \in \mathfrak{m}$ choose b with $0 < v(b) < v(a)$ and write $a = b(a/b)$. Every ideal of a valuation domain is a filtered union of principal ideals, hence $\mathfrak{m}$ is flat. The resolution $0 \rightarrow \mathfrak{m} \rightarrow V \rightarrow k \rightarrow 0$ therefore gives $\mathrm{Tor}_i^V(k, k) = 0$ for $i > 0$.

Let $I = 0 \times \mathfrak{m} \subset T$. Its image under q_1 is $\mathfrak{m}$, while $T/I \simeq V$ through q_0 . Tor-unitality is independent of the ambient ring for an ideal identified with an ideal elsewhere [14, Remark 1.9]; thus $q_0 : T \rightarrow V$ is Tor-unital. Tamme's excision theorem [15, Theorem 2] makes the K -theory square cartesian. Since q_0 is split by s , taking π_2 gives (27). $\square$

The same square is excisive for syntomic cohomology, giving all-degree Milnor excision without a relative presentation.

Proposition 6.2 (Syntomic excision on the fiber product). *For every $n \in \mathbf{Z}$ and $r \geq 1$, the square*

$$\begin{array}{ccc} \mathbf{Z}/p^r(n)_{\text{syn}}(T) & \longrightarrow & \mathbf{Z}/p^r(n)_{\text{syn}}(V) \\ \downarrow & & \downarrow \\ \mathbf{Z}/p^r(n)_{\text{syn}}(V) & \longrightarrow & \mathbf{Z}/p^r(n)_{\text{syn}}(k) \end{array}$$

is homotopy cartesian.

Proof. A homological epimorphism $R \rightarrow S$ satisfies $L_{S/R} = 0$: indeed $S \otimes_R^{\mathbf{L}} S \simeq S$, and cotangent base change with transitivity gives the claim. Thus the four cotangent exterior powers $\bigwedge^i L_{-/ \mathbf{Z}}$ form a cartesian square on the fiber product. Equivalently, for every T -complex M , tensoring the exact sequence $0 \rightarrow T \rightarrow V \oplus V \rightarrow k \rightarrow 0$ gives

$$M \longrightarrow (M \otimes_T^{\mathbf{L}} V) \oplus (M \otimes_T^{\mathbf{L}} V) \longrightarrow M \otimes_T^{\mathbf{L}} k.$$

The diffracted Hodge complex has a functorial conjugate filtration with graded pieces $(\bigwedge^i L_{R/\mathbf{Z}})_p^\wedge[-i]$, and Nygaard graded pieces are fibers of the Sen maps; see [2, Construction 4.7.1 and Remark 5.5.8]. Induction on finite conjugate steps gives cartesian squares of all Nygaard graded pieces. The finite Nygaard quotients are finite extensions of these pieces, hence are cartesian; so are their inverse limits, the Nygaard-completed steps. Syntomic cohomology is the fiber of divided Frobenius minus the canonical map [2, Proposition 7.4.6]; fibers preserve cartesian squares. Derived reduction modulo p^r is exact. $\square$

Corollary 6.3 (Milnor excision). *For every $n \geq 0$ and $r \geq 1$,*

$$\widehat{K}_n^M(T)/p^r \xrightarrow{\sim} \widehat{K}_n^M(V)/p^r \times_{\widehat{K}_n^M(k)/p^r} \widehat{K}_n^M(V)/p^r. \quad (28)$$

If a finite range $0 \leq n \leq N$ is fixed and $|k| > \max_{n \leq N} M_n$, the same holds with ordinary Milnor K -theory. In particular,

$$q_1 : \ker(K_n^M(T)/p \xrightarrow{q_0} A_n) \xrightarrow{\sim} J_n. \quad (29)$$

Proof. Lüders–Morrow prove that for every local p -henselian ring R , the syntomic symbol is an isomorphism

$$\widehat{K}_n^M(R)/p^r \xrightarrow{\sim} H^n(\mathbf{Z}/p^r(n)_{\text{syn}}(R)), \quad (30)$$

by [12, Theorem 3.1]. The rings V , k , and T are henselian: for T , a simple residue root lifts in each V -coordinate, and the lifts have the same residue. Moreover, $\mathbf{Z}/p^r(n)_{\text{syn}}(k) \in D^{\geq n}$ by the logarithmic de Rham–Witt description in characteristic p [6, Proposition 8.4 and Corollary 8.21]. Taking H^n of Proposition 6.2 gives (28). In a finite degree range, (20) replaces improved by ordinary Milnor K -groups. Finally, quotienting the Milnor presentation of V by the ideal generated by P gives the Milnor presentation of k : units and residue Steinberg relations lift. Hence $J_n = \ker(A_n \rightarrow K_n^M(k)/p)$, proving (29). $\square$

6.3 A universal degree-three class

The essential new operation lies in degree three; the higher operations will be forced by it.

Lemma 6.4 (Universal three-variable class). *Let R be a henselian local $\mathbf{Z}_p$ -algebra, and let $x, y, z \in R$ with $1 + xyz \in R^\times$. There is a natural class*

$$D_R(x; y, z) \in \widehat{K}_3^M(R)/p,$$

equal to $\{1 + xyz, y, z\}$ when y, z are units. It is functorial in R and satisfies

$$D(x; y, z) = -D(x; z, y), \quad (31)$$

$$D(x; y, z_1 z_2) = D(x z_2; y, z_1) + D(x z_1; y, z_2), \quad (32)$$

$$D(x; y, z h^p) = D(x h^p; y, z), \quad (33)$$

$$D(x; y, u) = \langle x u, y \rangle \{u\} \quad (u \in R^\times), \quad (34)$$

$$D(x; y, y) = D(-xy; y, -1). \quad (35)$$

In (34) use $K_2^M \simeq K_2$ when the residue field is large enough. All formulas include $p = 2$.

Proof. On $X = \text{Spec } \mathbf{Z}_p[X, Y, Z, (1 + XYZ)^{-1}]$ the rational Milnor class $\{1 + XYZ, Y, Z\}$ has zero residue at every codimension-one point. The only possible nonunit divisors are $Y = 0$ and $Z = 0$, where the first entry specializes to 1; the vertical divisor contributes no additional residue. The Gersten theorem for improved Milnor K -theory on smooth schemes over a complete DVR therefore gives a unique Nisnevich section extending the rational class [12, Theorem 4.1 and Corollary 4.4]. Pulling its henselian stalk to R defines D_R .

To justify specialization of the identities without using the endpoint theorem, choose for a given polynomial identity a smooth polynomial localization over $\mathbf{Z}_p$ in all variables, inverting exactly the expressions required to be units. At its generic point, the D -classes are ordinary Milnor symbols; the identity follows from multilinearity and Steinberg relations. Gersten injectivity [12, Corollary 4.4] makes the identity hold globally before specialization to R . In particular, this does not use injectivity of $\widehat{K}_3^M(R)/p$ into the Milnor group of the fraction field.

Formula (31) is skew symmetry; (32) expands the last Milnor entry. Before reduction modulo p the same calculation gives

$$D(x; y, z h^p) = D(x h^p; y, z) + p D(x z h^{p-1}; y, h),$$

hence (33). Equation (35) uses the integral identity $\{y, y\} = \{y, -1\}$, without dividing by 2.

For (34), the right side involves a Dennis–Stein class. Work first over a smooth local polynomial ring with residue field large enough for ordinary Milnor K_2 to agree with Quillen K_2 ; one can adjoin finitely many variables and specialize them to distinct residue classes if needed. The Dennis–Stein unit formula gives $\langle x u, y \rangle = \{1 + x y u, y\} \in K_2$. After multiplying by $\{u\}$ the two sides agree generically; Gersten injectivity makes them equal before specialization. Thus (34) holds on V without a relative presentation or endpoint injectivity. Everything is integral before reduction modulo p , so also holds for $p = 2$. $\square$

6.4 Relative degree-two and degree-three operations

We first identify relative K_2 with the principal-unit ideal in A_2 .

Lemma 6.5. *Assume $|k| > 5$ and ordinary and improved Milnor K -theory agree in degrees at most three. Then the natural map identifies*

$$K_2(V, \mathfrak{m})/p \xrightarrow{\sim} J_2 \subset A_2.$$

Proof. The coefficient sequence for relative K -theory reduces the claim to surjectivity of $K_3(V; \mathbf{F}_p) \rightarrow K_3(k; \mathbf{F}_p)$. Geisser–Levine identify $K_3(k; \mathbf{F}_p)$ with the image of Milnor symbols [9, Theorem 8.1]; their entries lift to units of V , proving surjectivity. The relative coefficient sequence therefore makes $K_2(V, \mathfrak{m}; \mathbf{F}_p) \rightarrow K_2(V; \mathbf{F}_p)$ injective. The universal-coefficient sequences contain natural injections of the integral groups modulo p into these coefficient groups, so $K_2(V, \mathfrak{m})/p \rightarrow K_2(V)/p$ is injective too.

By Matsumoto’s theorem, $K_2(k) = K_2^M(k)$; every residue symbol lifts to V , so $K_2(V) \rightarrow K_2(k)$ is surjective. Reducing the relative exact sequence modulo p identifies its image with $\ker(K_2(V)/p \rightarrow K_2(k)/p)$. Using (24), this is the kernel of $K_2^M(V)/p \rightarrow K_2^M(k)/p$, namely $PA_1 = J_2$. $\square$

For $g \in G$ and $w = \{1 + a\} \in P$ a principal unit, choose $\rho \in K^\times$ representing g with $0 < v(\rho) < v(a)$ and put

$$d_g(w) = \langle a/\rho, \rho \rangle \in J_2. \quad (36)$$

Lemma 2.2 supplies ρ .

Lemma 6.6. *Formula (36) induces a $\mathbf{F}_p$ -bilinear map $d : G \times P \rightarrow J_2$. If a unit $u \in V^\times$ represents a class of $A_1 \subset G$, then*

$$d_{\{u\}}(w) = w\{u\}. \quad (37)$$

Proof. The Dennis–Stein relations (21)–(23) give, for a common sufficiently small ρ ,

$$\left\langle \frac{a}{\rho}, \rho \right\rangle + \left\langle \frac{b}{\rho}, \rho \right\rangle = \left\langle \frac{a + b + ab}{\rho}, \rho \right\rangle,$$

so the construction is additive in the principal unit. For $\rho' = \rho h^p$ representing the same class in G , order valuations so that h is integral. Iterating (23) gives

$$\left\langle \frac{a}{\rho'}, \rho' \right\rangle = \left\langle \frac{a}{\rho}, \rho \right\rangle + p \left\langle \frac{a}{h}, h \right\rangle,$$

which is unchanged modulo p . Applying the same relation to a product of small representatives proves additivity in g . For $\rho = uh^p$ with small positive value, the unit-entry Dennis–Stein formula gives (37). $\square$

For the degree-three operation, if $a \in \mathfrak{m}$ define

$$f_g(a, b) = D(a/\rho; b, \rho), \quad (38)$$

and if $b \in \mathfrak{m}$ define

$$f_g(a, b) = -D(-b/\rho; -a, \rho), \quad (39)$$

where ρ is a sufficiently small positive representative of g .

Lemma 6.7. *The two formulas agree if $a, b \in \mathfrak{m}$, are independent of ρ , and are additive in g .*

Proof. Write $a = \rho A$ and $b = \rho B$. On the universal smooth source, the difference is the Milnor class $\{1 + \rho^2 AB, -\rho^2 AB, \rho\} = 0$; Lemma 6.4 extends it to specialization. Independence under $\rho \mapsto \rho h^p$ follows from (33), and additivity from (32) after choosing representatives with product sufficiently small. $\square$

Proposition 6.8 (Degree-three relative operation). *For every $g \in G$ there is an $\mathbf{F}_p$ -linear map $\Theta_g : J_2 \rightarrow A_3$, additive in g , such that for $w \in P$, $a \in A_1$ and $g, h \in G$,*

$$\Theta_g(wa) = -d_g(w)a, \quad (40)$$

$$\Theta_h d_g(w) = -\Theta_g d_h(w), \quad (41)$$

$$\Theta_g d_g(w) = d_g(w)e. \quad (42)$$

For $p = 2$, the e term in (42) is essential.

Proof. Construct a homomorphism on the absolute group $K_2(T)$. Write a Dennis–Stein symbol as $r = (x, x + a)$ and $t = (y, y + b)$, with $a, b \in \mathfrak{m}$. Put $U = 1 + xy$ and $H = 1 + x(y + b)$; both are units when the endpoint symbols are defined. Set

$$\Phi_g \langle r, t \rangle = f_g \left(x, \frac{b}{U} \right) + f_g \left(\frac{a}{H}, y + b \right). \quad (43)$$

For any finite relation, choose a representative ρ so small that it divides every coordinate difference. On a smooth universal localization, (43) has generic value

$$(\langle x + a, y + b \rangle - \langle x, y \rangle) \{\rho\}. \quad (44)$$

The usual Dennis–Stein first- and second-variable expansions give

$$\langle x + a, y + b \rangle - \langle x, y \rangle = \left\langle x, \frac{b}{1 + xy} \right\rangle + \left\langle \frac{a}{1 + x(y + b)}, y + b \right\rangle,$$

and Lemma 6.4 identifies multiplication by $\{\rho\}$ with the terms of (43).

We check the full Dennis–Stein presentation, including intermediate denominators. For a finite relation, take a single sufficiently small ρ and write

$$r = (x, x + \rho A), \quad t = (y, y + \rho B), \quad u = (z, z + \rho C).$$

Formula (43) is represented on the smooth polynomial localization in x, y, z, A, B, C, ρ obtained by inverting the endpoint and intermediate denominators. Each intermediate denominator has the same residue as its endpoint counterpart, so it specializes to a unit of V . The generic value is (44).

For (21), the generic image of $\langle r, t \rangle + \langle -t, -r \rangle$ is

$$(\langle x + \rho A, y + \rho B \rangle + \langle -y - \rho B, -x - \rho A \rangle - \langle x, y \rangle - \langle -y, -x \rangle) \{\rho\},$$

which vanishes by DS1 in the two coordinates. For (22), let $w = t + u + rtu$. The first coordinate is $w_0 = y + z + xyz$ and the second is $w_0 + \rho W$, where

$$W = B + C + Ayz + xBz + xyC + \rho(ABz + AyC + xBC) + \rho^2 ABC.$$

The generic image of $\langle r, t \rangle + \langle r, u \rangle - \langle r, w \rangle$ is DS2 for $(x + \rho A, y + \rho B, z + \rho C)$ minus DS2 for (x, y, z) , multiplied by $\{\rho\}$; hence it vanishes. For (23), the coordinate differences of tu , rt and ru are respectively

$$\rho(yC + zB + \rho BC), \quad \rho(xB + yA + \rho AB), \quad \rho(xC + zA + \rho AC).$$

The generic image of $\langle r, tu \rangle - \langle rt, u \rangle - \langle ru, t \rangle$ is the difference of the DS3 relations, multiplied by $\{\rho\}$, and is zero.

Each equality is one among universal D -classes on the chosen smooth localization. By Lemma 6.4, Gersten injectivity makes it valid before specialization, including zero entries or nonunits. The independence of f_g from the representative allows a common ρ for each finite relation. Thus (43) respects the absolute Dennis–Stein presentation and induces a homomorphism $\Phi_g : K_2(T) \rightarrow A_3$. It vanishes on $s_*K_2(V)$ since (43) is zero if $a = b = 0$.

Restrict to $\ker q_0$. Lemmas 6.1 and 6.5 identify this kernel modulo p with J_2 via q_1 ; the resulting map is Θ_g . For $w = \{1 + a\}$ and a unit u , the two evident lifts give

$$\Theta_g(w\{u\}) = D(a/(u\rho); u, \rho) = -\langle a/\rho, \rho \rangle \{u\},$$

using (31) and (34). This is (40). Choose representatives ρ, σ of g, h with $v(\rho) + v(\sigma) < v(a)$. The definitions give $\Theta_h d_g(w) = D(a/(\rho\sigma); \rho, \sigma)$, so (31) proves (41). If $g = h$, choose $2v(\rho) < v(a)$; then (35) and (34) yield

$$\Theta_g d_g(w) = D(a/\rho^2; \rho, \rho) = D(-a/\rho; \rho, -1) = d_g(w)e.$$

No division by 2 occurs. □

7 All-degree Milnor operations and the endpoint

The degree-three identities in Proposition 6.8 supply the coherence needed to extend the operation to every degree. We separate the formal graded-algebra argument from its valuation application.

7.1 A graph normal form

Let $A = \bigoplus_{m \geq 0} A_m$ be a connected $\mathbf{F}_p$ -graded-commutative algebra generated by A_1 . Fix $e \in A_1$ such that

$$u^2 = ue \quad (u \in A_1), \tag{45}$$

with $e = 0$ for odd p . Let $P \subset A_1$, $J = AP$, and let $G \supset A_1$ be an $\mathbf{F}_p$ -vector space. Choose a complement with ordered basis $(t_i)_{i \in I}$ and let C be the graded-commutative A -algebra generated by t_i subject to

$$t_i^2 = t_i e. \tag{46}$$

For an ordered finite subset $S = \{i_1 < \cdots < i_r\}$, put $t_S = t_{i_1} \cdots t_{i_r}$.

Lemma 7.1. *As a left or right A -module,*

$$C = \bigoplus_{S \subset I, |S| < \infty} At_S.$$

Consequently $JC = \bigoplus_S Jt_S$ and $A \cap JC = J$.

Proof. For odd p this is the squarefree basis of an exterior algebra. For $p = 2$, adjoining one generator is quotienting by the monic polynomial $T^2 - eT$, producing a free module with basis $1, T$. Adjoin finitely many variables successively, then pass to the filtered union. No condition that e be a nonzerodivisor is required. $\square$

Lemma 7.2 (Graph normal form). *Suppose that for every i there are degree-one linear maps $R_i : J_m \rightarrow J_{m+1}$ ($m \geq 1$) such that for homogeneous $q \in J$ and $a \in A$,*

$$R_i(qa) = (-1)^{\deg a} R_i(q)a, \quad (47)$$

$$R_j R_i(q) = -R_i R_j(q), \quad (48)$$

$$R_i^2(q) = R_i(q)e. \quad (49)$$

Put $d_i = R_i|_P$ and $B = C/(wt_i - d_i(w) : w \in P, i \in I)$. Then $A \rightarrow B$ is injective. For any graded vector-space splitting $A = J \oplus H$,

$$B \simeq A \oplus \bigoplus_{\emptyset \neq S \subset I} Ht_S \quad (50)$$

as graded vector spaces.

Proof. Give J its usual right A -module structure and prescribe right multiplication by t_i to be R_i . Equations (47)–(49) are exactly the relations needed for this right action to descend from the free associative extension to C . There is a right C -linear evaluation map

$$\epsilon : JC = J \otimes_A C \longrightarrow J, \quad q \otimes c \longmapsto q \cdot c,$$

restricting to the identity on J . Its kernel is generated by $qt_i - R_i(q)$, $q \in J$. Indeed, expand an element of JC in the squarefree basis. For a term $qt_{i_1} \cdots t_{i_r}$, successively subtract $(qt_{i_1} - R_{i_1}(q))t_{i_2} \cdots t_{i_r}$, then the analogous expression using $R_{i_1}(q)$ and t_{i_2} , and so forth. The resulting identity is

$$qt_S - R_{i_r} \cdots R_{i_1}(q) = qt_S - \epsilon(qt_S),$$

proving both containment directions for the kernel.

It is enough to take $q \in P$: equation (47) with graded commutativity implies $R_i(aq) = aR_i(q)$, and J is generated by P as an A -ideal. Thus the quotient ideal defining B is exactly $\ker \epsilon$. If $a \in A$ dies in B , then $a \in A \cap \ker \epsilon \subset A \cap JC = J$, but $\epsilon(a) = a$, hence $a = 0$. Splitting the coefficient of each nonconstant t_S into its J and H parts and evaluating the J part gives (50), including linear independence.

In degree n , only operations through degree $n - 1$ are needed. One can truncate A above degree n and set the top-degree operations to zero; omitted identities land outside the range. $\square$

7.2 Extending the degree-three operation

We next turn d_g and Θ_g into operations in all Milnor degrees.

Lemma 7.3 (Twisted derivation extension). *Let $M = T(E)/(Q)$ be a connected quadratic graded algebra with E in degree one. Let $f_0, f_1 : M \rightarrow A$ be graded algebra maps, and suppose $w_x = f_1(x) - f_0(x) \in P$ for $x \in E$. Suppose $d : P \rightarrow J_2$ and $\Theta : J_2 \rightarrow A_3$ satisfy*

$$\Theta(wa) = -d(w)a \quad (w \in P, a \in A_1). \quad (51)$$

Put $\delta(x) = d(w_x)$. Then

$$D(x_1 \cdots x_m) = \sum_{i=1}^m (-1)^{m-i} f_1(x_1) \cdots f_1(x_{i-1}) \cdot \delta(x_i) f_0(x_{i+1}) \cdots f_0(x_m) \quad (52)$$

defines $D : M_m \rightarrow J_{m+1}$. It satisfies

$$D(uv) = f_1(u)D(v) + (-1)^{\deg v} D(u)f_0(v). \quad (53)$$

If $s : A \rightarrow M$ is a section with $f_0s = f_1s = \text{id}$, then

$$D(s(a)) = 0, \quad D(zs(a)) = (-1)^{\deg a} D(z)a. \quad (54)$$

Proof. Formula (52) is multilinear on the tensor algebra. Separating the differentiated slot between words proves (53). For $x, y \in E$,

$$f_1(x)f_1(y) - f_0(x)f_0(y) = w_x f_0(y) + f_1(x)w_y \in J_2.$$

Using (51) and graded commutativity,

$$D(xy) = \Theta(f_1(x)f_1(y) - f_0(x)f_0(y)).$$

If $r \in Q$, both $f_0(r)$ and $f_1(r)$ vanish, so $D(r) = 0$. The Leibniz rule gives $D(urv) = 0$ for arbitrary words, proving descent to M . Every summand contains a factor $\delta(x_i) \in J_2$, hence lands in J . On diagonal degree-one generators $w_{s(a)} = 0$, so (53) gives (54). $\square$

Apply the lemma with

$$M = K_*^M(T)/p, \quad f_i = q_i, \quad s : K_*^M(V)/p \rightarrow K_*^M(T)/p, \quad (d, \Theta) = (d_g, \Theta_g).$$

Milnor K -theory modulo p has the required quadratic tensor-algebra presentation by Steinberg relations. For a unit pair $t_i = (u_i, u'_i) \in T^\times$, the operation is

$$\Phi_g^{(m)}\{t_1, \dots, t_m\} = \sum_{i=1}^m (-1)^{m-i} \{u'_1, \dots, u'_{i-1}\} d_g\{u'_i/u_i\} \{u_{i+1}, \dots, u_m\}. \quad (55)$$

It vanishes on diagonal classes and satisfies the module rule (54).

We explain why these operations descend to the *actual* ideal J , not just a presentation. Put $L_m = \ker(q_0 : K_m^M(T)/p \rightarrow A_m)$. The independent Milnor-excision isomorphism (29) says

$$q_1 : L_m \xrightarrow{\sim} J_m \quad (56)$$

in every fixed finite degree range. Thus each $\alpha \in J_m$ has a unique relative lift $\tilde{\alpha} \in L_m$. Restrict $\Phi_g^{(m)}$ to L_m and define

$$R_g(\alpha) = \Phi_g^{(m)}(\tilde{\alpha}) \in J_{m+1}. \quad (57)$$

Every summand of (55) contains $d_g(w_i) \in J_2$. The construction uses the genuine excision isomorphism (56) and automatically respects all relations in the actual ideal.

Proposition 7.4 (All-degree operations). *In every fixed finite degree range, the maps (57) are bilinear in g and satisfy*

$$R_g(w) = d_g(w) \quad (w \in P), \quad (58)$$

$$R_g(\alpha a) = (-1)^{\deg a} R_g(\alpha) a, \quad (59)$$

$$R_u(\alpha) = \alpha u \quad (u \in A_1), \quad (60)$$

$$R_h R_g(\alpha) = -R_g R_h(\alpha), \quad (61)$$

$$R_g^2(\alpha) = R_g(\alpha) e. \quad (62)$$

The identities include $p = 2$.

Proof. For $w = \{1 + a\} \in P$, the canonical relative lift is the class of $(1, 1 + a) \in T^\times$. Formula (55) in degree one gives (58). If $\tilde{\alpha} \in L_m$ lifts $\alpha \in J_m$, then $\tilde{\alpha} s_*(a)$ uniquely lifts αa ; (54) gives (59), an identity on the actual ideal.

Every element of J_m is a sum of wa , with $w \in P$ and $a \in A_{m-1}$. For $u \in A_1 \subset G$, (37) and (59) give

$$R_u(wa) = (-1)^{m-1} d_u(w) a = (-1)^{m-1} wua = wau,$$

by graded commutativity. This proves (60). For $wa \in J_2$, the module rule reads $R_g(wa) = -d_g(w)a$, which is exactly the defining formula (40) for Θ_g . Since these products span J_2 , $R_g|_{J_2} = \Theta_g$. The degree-three identities yield

$$R_h d_g(w) = -R_g d_h(w), \quad R_g d_g(w) = d_g(w) e. \quad (63)$$

These identities were obtained from the universal D calculation; no higher-degree injectivity is used.

For homogeneous $a \in A$, applying (59) twice to a generator $wa \in J$ gives

$$R_h R_g(wa) = R_h d_g(w) a, \quad R_g R_h(wa) = R_g d_h(w) a,$$

with identical total Koszul signs; (63) proves (61). Similarly $R_g^2(wa) = R_g d_g(w) a = d_g(w) ea = R_g(wa) e$. Since wa span J , this proves (62) in all degrees considered. For $p = 2$, the signs become plus signs but $R_g^2(-)e$ remains, and the class $e = \{-1\}$ cannot be omitted. $\square$

7.3 Endpoint injectivity

Theorem 7.5 (Milnor endpoint). *Let V be a complete rank-one valuation ring of mixed characteristic $(0, p)$. For every $n \geq 1$,*

$$\widehat{K}_n^M(V)/p \hookrightarrow K_n^M(\text{Frac } V)/p \quad (64)$$

is injective. Under the syntomic symbol and norm-residue isomorphism, this is injectivity of

$$H^n S_n(V) \longrightarrow H^n E_n(V). \quad (65)$$

Proof. First assume V nondiscrete and k sufficiently large. Fix $N \geq \max\{3, n\}$ so that the ordinary/improved comparison holds through degree N , and put $A = K_*^M(V)/p$ in that range. The inclusion $A_1 \hookrightarrow G$ is injective: if a unit is a p th power in K , its root has valuation zero and is a unit. Choose a complement $G = A_1 \oplus \bigoplus_i \mathbf{F}_p t_i$.

Relations (59), (61) and (62) satisfy Lemma 7.2; moreover $u^2 = ue$ in A_1 by (25). Thus there is a graded algebra B containing A injectively and generated by new degree-one classes t_i , with graph relations $wt_i = d_{t_i}(w)$, $t_i^2 = t_i e$. Every $g \in G$ acts as a degree-one generator with these relations.

We verify the field Steinberg relations $[x][1-x] = 0$ in B_2 . If $v(x) > 0$, choose ρ representing $[x] \in G$ with $0 < v(\rho) < v(x)$, and write $x = \rho h^p$. Since $1-x$ is a principal unit, the graph relation and Dennis–Stein identities give

$$[x][1-x] = d_{[x]}(1-x) = \langle -h^p, \rho \rangle = -\langle -\rho, h^p \rangle = -p\langle -\rho h^{p-1}, h \rangle = 0.$$

If $v(1-x) > 0$, interchange x and $1-x$. If both are units, their Steinberg relation holds in A_2 . If $v(x) < 0$, put $z = x^{-1}$; in G , $[1-x] = e + [x] + [1-z]$. The preceding unit case gives $[x][1-z] = -[z][1-z] = 0$, hence

$$[x][1-x] = [x]e + [x]^2 = 2[x]e = 0 :$$

for odd p , $e = 0$, while for $p = 2$, $2 = 0$. Thus the Milnor presentation of the field defines a graded map $K_*^M(K)/p \rightarrow B$ whose restriction to A_n is the injection $A_n \hookrightarrow B_n$. This proves (64) for large residue fields.

For finite k , only finitely many residue-cardinality bounds are needed through degree N . Choose a finite extension k'/k of degree d prime to p and large enough to meet them. Henselianity lifts it to a finite étale local valuation extension V'/V of the same degree. The value group is unchanged, so the nondiscrete argument holds over V' . If $x \in H^n S_n(V)$ dies generically, its restriction to V' dies generically and is zero by the large-residue endpoint. The finite étale syntomic norm of [12, Lemma 3.3 and Corollary 3.4] obeys

$$\mathrm{Nm}_{V'/V} \circ \mathrm{res}_{V'/V} = d$$

on syntomic cohomology. Since d is invertible modulo p , restriction is injective and $x = 0$.

If V is discrete, it is a complete DVR and the endpoint is covered by the regular comparison theorem [4, Theorem 4.15 and Proposition 5.2]. Finally, (30) identifies the source of (64) with $H^n S_n(V)$, and norm-residue identifies $K_n^M(K)/p$ with $H_{\mathrm{ét}}^n(K, \mu_p^{\otimes n})$ [18, Theorem 6.16]; both maps agree with the generic Galois symbol, proving (65). $\square$

8 Frobenius base change and twist periodicity

The remaining obstruction is the family of completion defects E_k . We show that after one conservative Frobenius-type base change, all twists become equivalent. The geometry of the Hodge–Tate stack is used here. Put

$$A = V/p, \quad \psi : V \longrightarrow A, \quad v \longmapsto \bar{v}^p, \quad \psi^*(-) = - \otimes_{V, \psi}^{\mathbf{L}} A.$$

Throughout the section V is nondiscrete, complete and rank one.

8.1 Finite-Tor base change

Lemma 8.1 (Eventual-stabilization base change). *Let B have Tor amplitude $[-d, 0]$ over R , and let (X_i) be an inverse system such that for each i the fibers $\mathrm{fib}(X_j \rightarrow X_i)$ for $j \geq i$ lie in $D^{\geq c_i}$, where $c_i \rightarrow \infty$. Then*

$$B \otimes_R^{\mathbf{L}} \varprojlim_i X_i \xrightarrow{\sim} \varprojlim_i (B \otimes_R^{\mathbf{L}} X_i).$$

The same applies to the partial-totalization tower of a cosimplicial object whose terms have a uniform lower bound.

Proof. Put $\widehat{X} = \varprojlim_i X_i$. Taking the inverse limit in the fiber triangles over $j \geq i$ yields $\text{fib}(\widehat{X} \rightarrow X_i) \in D^{\geq c_i}$. Tensoring with B lowers connectivity by at most d ; the same estimate applies to the inverse limit of the base-changed tower. Hence the canonical map is an isomorphism in any fixed cohomological degree for $i \gg 0$. For totalizations, normalized cosimplicial degree r is a summand of the r th term and appears with a shift $[-r]$. Thus successive partial-totalization fibers become arbitrarily connective, and the inverse-system argument applies. $\square$

Lemma 8.2 (Conservativity and one-degree detection). *The Frobenius pullback $F_A^* : D(A) \rightarrow D(A)$ is conservative; hence $\psi^* : D(V) \rightarrow D(A)$ is conservative on derived p -complete complexes. The V -module A via ψ has Tor amplitude $[-1, 0]$. Moreover, if a V -module M is annihilated by a power of p and*

$$\text{Tor}_1^V(M, A; \psi) = 0,$$

then $M = 0$.

Proof. A valuation ring has weak dimension at most one, giving the Tor-amplitude assertion. Extend the valuation to $K(p^{1/p})$ and let V' be its valuation ring. It is faithfully flat over V . Take $\pi \in V'$ with $\pi^p = p$, and put $A' = V'/p$. Frobenius on A' factors as

$$A' \longrightarrow V'/\pi \longrightarrow A'.$$

The first arrow has nilpotent kernel. The second, induced by $x \mapsto x^p$, is faithfully flat: V'/π is a quotient of a valuation ring, its finitely generated ideals are principal, and the annihilator criterion applies as in [13, Proposition A.4 and Corollary A.5]; see also [10]. Derived base change by a nilpotent quotient is conservative, so $F_{A'}^*$ is conservative; faithfully flat descent along $A \rightarrow A'$ proves the assertion for F_A^* .

For derived p -complete N , $\psi^*N \simeq F_A^*(N \otimes_V^{\mathbf{L}} A)$. Thus $\psi^*N = 0$ implies $N/p = 0$, hence $N = 0$ by derived p -completeness.

For the last statement, base change to V' and factor ψ' through $B = V'/\pi$ and then the faithfully flat map $B \rightarrow A'$. The two-term resolution of B over V' gives

$$\text{Tor}_1^{V'}(M \otimes_V V', A'; \psi') \simeq (M \otimes_V V')[\pi] \otimes_B A'.$$

Its vanishing implies that multiplication by π is injective on $M \otimes_V V'$. But $p^N M = 0$ for some N , so π^{p^N} annihilates the base change. An injective nilpotent endomorphism forces it to vanish, and faithful flatness gives $M = 0$. $\square$

8.2 The actual twisted Nygaard filtrations

We use the multiplicative induced twisted Nygaard filtration of [4, Constructions 3.1 and 3.3 and Remark 3.4] on the *actual filtered complexes*, not merely on their graded pieces. For every $k \in \mathbb{Z}$,

$$F^0 H_k = H_k, \quad F^r H_k \otimes F^s H_\ell \longrightarrow F^{r+s} H_{k+\ell}. \quad (66)$$

All-integer twists and multiplicativity follow from filtered Breuil–Kisin twists [2, Section 5.5]. On QRSP charts this filtration is induced by images of Nygaard ideals in Hodge–Tate cohomology, with finite quotients B_q^k as in Section 4.

Lemma 8.3 (Completion commutes with ψ^*). *For every k ,*

$$\psi^* \widehat{H}_k \xrightarrow{\sim} \varprojlim_q \psi^* B_q^k.$$

Consequently $\psi^ E_k$ is the completion defect of the base-changed filtered object $\psi^* H_k$.*

Proof. By Theorem 3.6, the twisted Bott graded fibers become increasingly connective. The induced filtration satisfies $\mathrm{gr}_F^j H_k \in D^{\geq j-p+1}$ for $j \geq p$, uniformly in k ; this is the form of weak F -smoothness used in [4, Remark 4.5]. Thus for $q' \geq q \geq p$,

$$\mathrm{fib}(B_{q'}^k \rightarrow B_q^k) \in D^{\geq q-p+1}.$$

The V -module A via ψ has Tor amplitude $[-1, 0]$ by Lemma 8.2. Apply Lemma 8.1; exactness of ψ^* identifies the completion cofibers. $\square$

If a twist-one class becomes a unit after ψ base change, (66) makes it a filtered unit.

Lemma 8.4 (Filtered unit). *Suppose $\eta \in H^0(\psi^* H_1)$ and $\eta^{-1} \in H^0(\psi^* H_{-1})$ multiply to 1. Multiplication by η gives inverse equivalences*

$$\psi^*(F^\bullet H_k) \xrightarrow{\sim} \psi^*(F^\bullet H_{k+1})$$

for every k , and equivalences of their completion defects.

Proof. Both classes are in filtration zero because $F^0 H_{\pm 1} = H_{\pm 1}$. Multiplicativity gives maps on every filtration step, whose products are the identity in $H^0(F^0 H_0)$. Thus the two filtered multipliers are inverses in the filtered derived category. Finite quotients, inverse limits, and cofibers preserve equivalences, and Lemma 8.3 identifies the defects with $\psi^* E_k$. $\square$

8.3 A geometric trivialization of the Breuil–Kisin line

We construct the required unit. The quotient presentation of Hodge–Tate prismatization for a regular local prism in [3, Construction 9.4 and Proposition 9.5] is the geometric input. For a bounded prism $(P, (E))$ presenting a complete regular local ring, the Hodge–Tate Cartier–Witt stack is a quotient of a $G_a^\sharp$ -torsor by the semidirect product of the divided-power tangent group and $G_m^\sharp$, and the $G_m^\sharp$ character on the Breuil–Kisin line is the zeroth Witt coordinate; compare [2, Section 4.4].

Lemma 8.5. *There exist $a, b \in \mathfrak{m}_V$ with*

$$a^p b = p, \quad a^{p^2} \in pV.$$

For

$$P = \mathbf{Z}_p[[T, B]], \quad E = p - T^p B, \quad R = P/(E),$$

with $\delta(T) = \delta(B) = 0$, the pair $(P, (E))$ is a bounded prism and R is a p -torsion-free complete regular local ring of dimension two. Evaluation $T \mapsto a, B \mapsto b$ defines a continuous map $R \rightarrow V$.

Proof. Normalize $v(p) = 1$. By density choose $p^{-2} \leq v(a) < p^{-1}$ and put $b = p/a^p$. Then $a, b \in \mathfrak{m}_V$, $a^p b = p$ and $a^{p^2} \in pV$. The ring P is regular local of dimension three, and $E \in (p, T, B) \setminus (p, T, B)^2$, so R is regular local of dimension two. Modulo p , $E = -T^p B$ is a nonzerodivisor in $\mathbf{F}_p[[T, B]]$, proving p -torsion-freeness of R . Finally $\delta(E)$ has unit constant term $\delta(p) = 1 - p^{p-1}$; hence $p \in (E, \varphi(E))$. Completeness and boundedness are immediate, and the evaluation converges because $v(a), v(b) > 0$. $\square$

Lemma 8.6 (Hodge–Tate global sections after ψ base change). *Let $X = \mathrm{Spf} V$ and X^{HT} be its absolute Hodge–Tate stack. Put*

$$Z = \mathrm{Spec} A \times_{X, \psi} X^{\mathrm{HT}},$$

with derived fiber product. For every $k \in \mathbf{Z}$, the sections of the pulled k th Breuil–Kisin line identify multiplicatively with

$$R\Gamma(Z, \mathcal{O}\{k\}) \simeq \psi^* \bar{\Delta}_V \{k\}. \quad (67)$$

Consequently, a trivialization of the line and its dual on Z gives inverse classes after canonical derived reduction $\psi^ \bar{\Delta}_V \{\pm 1\} \rightarrow \psi^* H_{\pm 1}$.*

Proof. Choose the perfectoid valuation cover $V \rightarrow W$ of Lemma 2.3 and set $R_a = (W^{\otimes_V (a+1)})_p^\wedge$. These are p -torsion-free QRSP rings. By prismatic comparison and flat-local descent [3, Lemma 6.3, Theorem 7.17, Corollary 7.18 and Corollary 8.17], X^{HT} is covered by derived formal affines with rings $\bar{\Delta}_{R_a}$, whose cosimplicial object is the full Čech nerve. The k th Breuil–Kisin line has a discrete module of sections on each QRSP chart. These section complexes have a common lower bound, and descent identifies the totalization with $\bar{\Delta}_V \{k\}$.

After derived base change by ψ , the charts remain derived affine and the local sections become $\psi^* \bar{\Delta}_{R_a} \{k\}$. The only issue is commuting ψ^* with the infinite totalization. Normalized cosimplicial degree a is a summand of the a th term and appears with shift $[-a]$; as the unbased terms are uniformly connective, partial-totalization fibers become arbitrarily connective. Since ψ^* has Tor amplitude $[-1, 0]$ by Lemma 8.2, the totalization case of Lemma 8.1 applies. Flat-local descent gives (67). The identifications come from twist-graded multiplication and are multiplicative in k .

There is a derived distinction: (67) computes the pullback of the integral Hodge–Tate line. By definition,

$$\psi^* H_k = \psi^* (\bar{\Delta}_V \{k\} / p) \simeq \psi^* \bar{\Delta}_V \{k\} \otimes_{\mathbf{Z}_p}^{\mathbf{L}} \mathbf{F}_p.$$

Even though the scalar map lands in characteristic p , derived reduction can retain a Tor term. The reduction map is multiplicative, so inverse integral line sections remain inverse on $\psi^* H_{\pm 1}$. $\square$

Proposition 8.7 (Frobenius-pulled Breuil–Kisin-line unit). *There is a unit $\eta \in H^0(\psi^* H_1)$ with inverse in $H^0(\psi^* H_{-1})$. Consequently,*

$$\psi^* E_k \simeq \psi^* E_\ell \quad (k, \ell \in \mathbf{Z}). \quad (68)$$

Proof. Let $f : \mathrm{Spec} A \rightarrow \mathrm{Spf} R$ be given by $T \mapsto a^p \bmod p$, $B \mapsto b^p \bmod p$. Consider the base change of the Hodge–Tate Cartier–Witt stack of R along f . Its quotient presentation is an equivalence over $\mathrm{Spf} R$, and the torsor and the acting divided-power groups are flat over the base, so termwise base change computes the derived base change. For an A -algebra S , the adjoint δ -map $P \rightarrow W(S)$ sends

$$T \mapsto [a^p], \quad B \mapsto [b^p].$$

Since $a^{p^2} = 0$ in A , it sends E to p . The pulled Cartier–Witt divisor is the standard divisor $p : W(S) \rightarrow W(S)$. The torsor of splittings in [3, Proposition 9.5] is therefore

$$\{\tau \in W(S) : p\tau = p\} = 1 + W(S)[F] = G_m^\sharp(S).$$

The divided-power tangent action is controlled by

$$dE = -pT^{p-1}B dT - T^p dB.$$

Both coefficients vanish after applying f : the first because $p = 0$, the second because $a^{p^2} = 0$ in A . Thus the tangent group acts trivially. If y is the zeroth Witt coordinate of τ , it is invertible and $y(u\tau) = u_0 y(\tau)$ for $u \in G_m^\sharp$. Hence y is a nowhere-vanishing equivariant section of the character defining the Breuil–Kisin line (or its dual, according to sign convention), trivializing it after f .

Pull the trivialization along $R \rightarrow V$ to $Z = \operatorname{Spec} A \times_{\operatorname{Spf} V, \psi} (\operatorname{Spf} V)^{\operatorname{HT}}$. By Lemma 8.6, its twist-graded sections are the derived ψ -pullbacks of the integral Hodge–Tate Breuil–Kisin lines, compatibly with multiplication. Thus there are inverse sections in $H^0 \psi^*(\bar{\Delta}_V \{1\})$ and $H^0 \psi^*(\bar{\Delta}_V \{-1\})$. Transport them through multiplicative derived reduction to $\psi^* H_{\pm 1}$; this step is essential because reduction can retain a Tor term. Now apply Lemmas 8.3 and 8.4. Multiplication by η identifies adjacent filtered twists and their defects; iteration gives (68). $\square$

9 Vanishing of the Nygaard-completion defects

We combine the independent endpoint theorem (Theorem 7.5) with twist periodicity (Proposition 8.7).

Theorem 9.1 (Defect vanishing). *Let V be a nondiscrete complete rank-one valuation ring of mixed characteristic. Then $E_k(V) = 0$ for every $k \in \mathbf{Z}$.*

Proof. Proposition 5.6 gives $E_k \in D^{\geq 1}$ for all k . We prove inductively

$$E_k \in D^{\geq s} \quad \text{for all } k \quad (69)$$

for every $s \geq 1$. Assume (69). Fix a weight $j \geq s+2$. By Proposition 4.4, $H^s C_j \simeq H^{s-1} E_{j+p-1} = 0$. Thus the Bott map

$$H^{s+1} S_j(V) \longrightarrow H^{s+1} S_{j+p-1}(V) \quad (70)$$

is injective; the same holds at all later weights. Bhatt–Mathew’s Bott localization theorem identifies the telescope with generic étale cohomology:

$$\operatorname{colim}_{m \geq 0} S_{j+m(p-1)}(V) \xrightarrow{\sim} E_j(V), \quad (71)$$

with the usual coefficient identifications [4, Theorem 5.1]. Since all transition maps on H^{s+1} are injective, the comparison $H^{s+1} S_j(V) \rightarrow H^{s+1} E_j(V)$ is injective. It is surjective by Proposition 2.6, since $s+1 < j$. Thus it is an isomorphism for every $j \geq s+2$.

Take $j = s+2$ and $k = j+p-1$. The Bott map on H^{s+1} is an isomorphism because both comparison maps are isomorphisms in that degree and the generic Bott class is invertible. On H^{s+2} , Theorem 7.5 makes the weight- $(s+2)$ comparison injective. Naturality with the generic Bott class makes $H^{s+2} S_j(V) \rightarrow H^{s+2} S_k(V)$ injective. The long exact cohomology sequence of C_j gives $H^{s+1} C_j = 0$. Apply Proposition 4.4 with $q = j = s+2$:

$$H^s E_{s+p+1} = 0. \quad (72)$$

This defect belongs to $D^{\geq s+1}$.

Pull back by ψ . Since its Tor amplitude is $[-1, 0]$, (72) implies $\psi^* E_{s+p+1} \in D^{\geq s}$. By (68), $\psi^* E_k$ has this same bound for every k . From the induction hypothesis and the one-degree Tor spectral sequence,

$$H^{s-1}(\psi^* E_k) \simeq \operatorname{Tor}_1^V(H^s E_k, A; \psi) = 0.$$

The module $H^s E_k$ is annihilated by p . Lemma 8.2 forces $H^s E_k = 0$. Therefore $E_k \in D^{\geq s+1}$ for every k , closing the induction. Each E_k is arbitrarily connective and thus zero. $\square$

Remark 9.2 (No circularity). The endpoint theorem does not use Nygaard completeness or defect vanishing: it uses Tor-unital excision, the universal degree-three class and the all-degree Milnor normal form. Conversely, weak F -smoothness and separation do not use endpoint injectivity. They meet only at (72) in the induction above.

10 Universal F -smoothness of valuation rings

The rank-one results also imply that all valuation rings are F -smooth. We extend the weak F -smoothness and Nygaard-completeness statements separately, using the valuation-theoretic excision square. Theorem 3.6 proves weak F -smoothness for complete rank-one valuation rings, while Theorem 9.1 proves the vanishing of their twisted Nygaard-completion defects. The passage to arbitrary rank requires, in addition to excision, a uniform Tor-amplitude estimate. We give the argument below.

10.1 The rank-one and characteristic- p cases

For a p -quasisyntomic ring R let

$$G_i(R) = \mathrm{gr}_N^i \Delta_R/p, \quad W_i(R) = \mathrm{fib}(\theta : G_i(R) \longrightarrow G_{i+p}(R)) \quad (i \geq 0).$$

As in Section 4.2, put

$$H_k(R) = \overline{\Delta}_R\{k\}/p, \quad \widehat{H}_k(R) = \varprojlim_q B_q^k(R), \quad E_k(R) = \mathrm{cofib}(H_k(R) \longrightarrow \widehat{H}_k(R)).$$

Here $B_q^k(R)$ denotes the finite quotient for the induced twisted Nygaard filtration of Bhatt–Mathew [4, Construction 3.3]; this is precisely the intrinsic finite quotient used in Section 4.2, not a separatedness surrogate.

We use the following form of the Bhatt–Mathew criterion [4, Proposition 4.3 and Proposition 2.6]: if R is weakly F -smooth and $E_k(R) = 0$ for all $k \in \mathbf{Z}$, then R is F -smooth. Conversely, F -smoothness forces the vanishing of all these completion defects.

Proposition 10.1. *Every p -complete mixed-characteristic rank-one valuation ring is F -smooth. Every characteristic- p valuation ring is F -smooth. In either case $E_k = 0$ for all $k \in \mathbf{Z}$.*

Proof. For a nondiscrete rank-one ring, weak F -smoothness is Theorem 3.6 and $E_k = 0$ is Theorem 9.1, so the preceding criterion applies. A discrete p -complete rank-one ring is a complete DVR and hence F -smooth by [4, Theorem 4.15].

A characteristic- p valuation ring is Cartier smooth by the results of Gabber–Ramero and Gabber, as recorded in Kelly–Morrow, *K-theory of valuation rings*, §2 (Compos. Math. **157** (2021), 1121–1142). It is therefore F -smooth by [4, Proposition 4.14]. In both cases the completion defects vanish by [4, Proposition 4.3]. $\square$

10.2 Excision for the weak Frobenius fibers and completion defects

Let V be a p -complete valuation ring of mixed characteristic $(0, p)$, and set

$$\mathfrak{q} = \sqrt{pV}, \quad B = V_{\mathfrak{q}}, \quad C = V/\mathfrak{q}, \quad D = \kappa(\mathfrak{q}). \quad (73)$$

We consider the valuation-theoretic Milnor square

$$\begin{array}{ccc} V & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & D. \end{array} \quad (74)$$

By [1, Remark 7.24] these are p -complete valuation rings, and [1, Proposition 7.25] proves excision for the Nygaard-complete prismatic pieces and steps on this square.

Lemma 10.2 (Excision for weak fibers and completion defects). *For every $i \geq 0$ and $k \in \mathbf{Z}$, each of the functors*

$$W_i(-), \quad H_k(-), \quad B_q^k(-), \quad \widehat{H}_k(-), \quad E_k(-)$$

carries the square (74) to a cartesian square (for all $q \geq p$).

Proof. The weak fibers. The excision of the Nygaard graded pieces follows from [1, Proposition 7.25]: Nygaard completion does not alter a fixed finite associated graded piece, and reduction modulo p is exact. Thus G_i is excisive. Multiplication by θ is natural, so its fiber W_i also satisfies excision.

The Hodge–Tate twists before Nygaard completion. By [1, Lemma 7.22], all relevant derived exterior powers of the cotangent complex satisfy excision on (74). They are the associated graded terms of the conjugate filtration on the p -complete diffracted Hodge complex $\widehat{\Omega}_R^D$ of Bhatt–Lurie [2, Construction 4.7.1]. Finite conjugate stages therefore satisfy excision. After reduction modulo p , the conjugate filtration is exhaustive; filtered colimits commute with finite limits in the stable category. Consequently $\widehat{\Omega}_R^D/p$ satisfies excision. The Sen eigenspace formula [2, Remark 4.7.5] gives, with the standard sign convention,

$$\overline{\Delta}_R\{k\} \simeq \text{fib}(\Theta + k : \widehat{\Omega}_R^D \longrightarrow \widehat{\Omega}_R^D).$$

Taking fibers and derived reduction modulo p preserves cartesian squares, proving excision for H_k .

Finite quotients and defects. The finite associated graded pieces of the actual induced twisted Nygaard filtration admit the functorial description [4, Construction 3.3]

$$\text{gr}_N^r H_k(R) \simeq \text{cofib}(\theta : G_{r-p}(R) \longrightarrow G_r(R)), \quad G_j = 0 \ (j < 0). \quad (75)$$

By the first paragraph, these pieces satisfy excision. Each B_q^k is assembled from finitely many of them by extensions, so finite quotients satisfy excision. Limits of cartesian squares are cartesian, hence $\widehat{H}_k = \text{R}\varprojlim_q B_q^k$ is excisive. Finally E_k is the cofiber of the natural transformation $H_k \rightarrow \widehat{H}_k$, and finite limits and colimits agree in the stable category. This proves the lemma. $\square$

10.3 Detecting Tor-amplitude over a valuation ring

The following statement is tailored to the mod- p complexes W_i . It avoids assuming that a lower-connective subcategory is closed under fibers (which it is not in general).

Lemma 10.3 (Localization and Tor-amplitude over a valuation domain). *Let V be a valuation domain with p a nonzero nonunit, let $S \subset V \setminus pV$ be a multiplicative subset, and let $M \in D(V/p)$. Fix an integer a and assume:*

1. $S^{-1}M$ has p -complete Tor-amplitude in degrees $\geq a$ over $S^{-1}V$;

2. for every $f \in S$, the complex $M \otimes_V^{\mathbf{L}} V/f$ belongs to $D^{\geq a}$.

Then M has p -complete V -Tor-amplitude in degrees $\geq a$.

Proof. Write $V_S = S^{-1}V$. Since M_S comes from $D(V_S/p)$ and p is a nonzerodivisor in V_S , multiplication by p on M_S is nullhomotopic. Consequently

$$M_S \otimes_{V_S}^{\mathbf{L}} V_S/p \simeq M_S \oplus M_S[1].$$

The first hypothesis, tested on the discrete module V_S/p , therefore forces $M_S \in D^{\geq a+1}$.

For each $f \in S$, the element f is a nonzerodivisor in V . Hence $M \otimes_V^{\mathbf{L}} V/f$ is the cofiber of multiplication by f on M . The second hypothesis and the associated long exact sequence imply that multiplication by f is an isomorphism on $H^t(M)$ for every $t < a$ and an injection on $H^a(M)$. Meanwhile $S^{-1}H^t(M) = H^t(M_S) = 0$ for every $t \leq a$. Thus every class in $H^t(M)$, $t \leq a$, is killed by some $f \in S$; since multiplication by every such f is injective, the class is zero. We conclude that $M \in D^{\geq a+1}$.

Finally, every module over a valuation domain has flat dimension at most one. If N is any discrete V/p -module, resolve it as a V -module by two flat terms. Tensoring that resolution with $M \in D^{\geq a+1}$ gives $M \otimes_V^{\mathbf{L}} N \in D^{\geq a}$. This is the required p -complete Tor-amplitude bound. $\square$

10.4 Valuation rings of arbitrary rank

Proposition 10.4. *Every p -complete mixed-characteristic valuation ring V , of arbitrary rank and value group, is F -smooth.*

Proof. The valuation-ring cotangent estimate of Gabber–Ramero, used in [1, Proposition 7.25], makes V p -quasisyntomic. Set $\mathfrak{q}, B, C, D$ as in (73). Because V is p -adically separated, $\mathfrak{q}$ is its smallest nonzero prime. Indeed, if $\mathfrak{r} \neq (0)$ is prime and $0 \neq x \in \mathfrak{r}$, choose n with $x \notin p^n V$. Comparability of principal ideals gives $p^n V \subset xV$, hence $p \in \mathfrak{r}$ and $\mathfrak{q} \subset \mathfrak{r}$. Thus $\mathfrak{q}$ has height one. Consequently $B = V_{\mathfrak{q}}$ is a p -complete rank-one valuation ring. The rings C and D are valuation rings of characteristic p . All three are F -smooth by Proposition 10.1.

Weak F -smoothness. Fix $i \geq 0$ and put

$$M = W_i(V), \quad a = i + 1, \quad S = V \setminus \mathfrak{q}.$$

The localization is relatively perfect modulo p : localization at S and at S^p gives the same characteristic- p algebra. Hence [4, Proposition 3.7] gives

$$S^{-1}M \simeq W_i(B). \tag{76}$$

Since B is F -smooth, the right side has p -complete B -Tor amplitude in degrees $\geq i + 1$.

Let $f \in S$. By Lemma 10.2, the valuation square gives a fiber sequence

$$M \longrightarrow W_i(B) \oplus W_i(C) \longrightarrow W_i(D).$$

The element f is invertible in B and in D , so tensoring with V/f yields

$$M \otimes_V^{\mathbf{L}} V/f \simeq W_i(C) \otimes_V^{\mathbf{L}} V/f.$$

For any $x \in \mathfrak{q}$, comparability of xV with fV and $f \notin \mathfrak{q}$ gives $xV \subset fV$; therefore $\mathfrak{q} \subset fV$ and $V/f \simeq C/\bar{f}$. Moreover, $\bar{f} \neq 0$ is a nonzerodivisor in the valuation domain C , so the two-term resolution of V/f identifies $C \otimes_V^{\mathbf{L}} V/f \simeq C/\bar{f}$. We obtain

$$M \otimes_V^{\mathbf{L}} V/f \simeq W_i(C) \otimes_C^{\mathbf{L}} C/\bar{f} \in D^{\geq i+1}, \quad (77)$$

where the last bound follows from the weak F -smoothness of C . Apply Lemma 10.3 to (76) and (77). It follows that $W_i(V)$ has p -complete V -Tor-amplitude in degrees $\geq i+1$. As i was arbitrary, V is weakly F -smooth.

Nygaard completeness. For any $k \in \mathbf{Z}$, Lemma 10.2 gives a fiber sequence

$$E_k(V) \longrightarrow E_k(B) \oplus E_k(C) \longrightarrow E_k(D).$$

The three terms on the right vanish by Proposition 10.1 and the Bhatt–Mathew criterion. Thus $E_k(V) = 0$ for every k . Weak F -smoothness together with vanishing of the completion defects proves the claim by [4, Proposition 4.3]. $\square$

10.5 Removing the p -completeness hypothesis

Lemma 10.5. *Let V be a valuation ring in which p is a nonzero nonunit. Its ordinary p -adic completion $\widehat{V} = \varprojlim_n V/p^n$ is a p -torsion-free, p -complete valuation ring.*

Proof. The inverse limit of the local rings V/p^n is local. We first verify

$$\ker(\widehat{V} \longrightarrow V/p^n) = p^n \widehat{V}. \quad (78)$$

Indeed, a compatible system in the kernel can be divided modulo p^m ($m \geq n$) by p^n to obtain a unique element modulo p^{m-n} , because p is a nonzerodivisor in V . The divisions are compatible and define an element of $\widehat{V}$. The opposite inclusion is immediate.

Take $0 \neq x \in \widehat{V}$. Choose n and $a \in V \setminus p^n V$ representing the nonzero image of x in V/p^n . Since principal ideals are comparable, $p^n V \subset aV$, so $p^n = ac$ for some nonunit $c \in V$. By (78),

$$x = a + p^n z = a(1 + cz), \quad z \in \widehat{V}.$$

The element c belongs to the maximal ideal of $\widehat{V}$, hence $1 + cz$ is a unit. Every nonzero element of $\widehat{V}$ is therefore associated to the image of an element of V , which shows that its nonzero principal ideals are linearly ordered.

These elements are not zero divisors. Indeed, any nonzero image $a \in V$ divides some p^N , and similarly $b \mid p^M$. If ab had zero image in $\widehat{V}$, it would belong to every $p^r V$, contradicting $ab \mid p^{N+M}$ (take $r = N + M + 1$). Thus $\widehat{V}$ is a valuation domain. Multiplication by p is injective, and p -completeness is built into its construction. $\square$

Proof of Theorem 1.2. If p is invertible in V , its derived p -completion is zero and the F -smoothness condition is vacuous. If V has characteristic p , use Proposition 10.1. Otherwise V is a valuation domain of mixed characteristic $(0, p)$ with p a nonunit. Since V is p -torsion-free, its ordinary p -adic completion agrees with its derived p -completion. By Lemma 10.5 this completion is again a valuation ring, and by Proposition 10.4 it is F -smooth. The valuation-ring cotangent estimate gives the required p -quasisyntomicity of V . Finally, as observed in [4, Definition 4.1], F -smoothness depends only on the p -completion. Hence V is F -smooth. This holds for every prime p . $\square$

Remark 10.6. The proof of Theorem 1.2 uses the all-weight comparison of Theorem 1.1 only through the rank-one inputs of Theorems 3.6 and 9.1; it does not follow formally from the arbitrary-rank comparison statement. In particular, the excision of E_k and the Tor bound for W_i are separate requirements.

11 Proof of the all-weight comparison

Proposition 11.1 (Complete rank-one case). *For every complete rank-one valuation ring V of mixed characteristic $(0, p)$ and every $n \geq 0$,*

$$\mathrm{cofib}(S_n(V) \longrightarrow E_n(V)) \in D^{\geq n}(\mathbf{F}_p).$$

Proof. Suppose first that V is nondiscrete and $n \geq 1$. By Theorem 9.1, all completion defects vanish. Proposition 4.4 therefore gives

$$\tau_{\leq j-1} C_j(V) = 0 \quad (j \geq 1).$$

It follows that multiplication by v_1 is injective on $H^q S_j(V)$ for $q \leq j$. Applying this to the successive maps in the Bott tower starting in weight n , and using the localization equivalence (71), shows that

$$H^q S_n(V) \longrightarrow H^q E_n(V)$$

is injective for $q \leq n$. Proposition 2.6 gives surjectivity for $0 \leq q < n$. For $q < 0$, the target vanishes and injectivity forces the source to vanish as well. Thus the comparison is an isomorphism below degree n and injective in degree n .

If V is discrete, it is a complete discrete valuation ring, and the assertion follows from the comparison theorem for regular rings [4, Theorem 4.15 and Proposition 5.2]. For $n = 0$, both complexes are connective and the comparison induces the identity on $H^0 = \mathbf{F}_p$; see [2, Section 8.4]. This proves the proposition. $\square$

Proof of Theorem 1.1. Proposition 11.1 establishes the claim in complete rank one. The valuative excision argument of Bachmann–Elmanto–Morrow, stated as Proposition 2.1, then gives the result for every henselian valuation ring of mixed characteristic. No assumption on the rank, value group, completeness, or prime p is required. $\square$

12 Applications

We record consequences of Theorems 1.1 and 1.2 that follow by applying existing comparison, descent, and stability results. The prismatic completion statements used in proving Theorem 1.2 are not restated here. Throughout this section, $\mathbf{Z}(i)_{\mathrm{mot}}$ denotes the (not generally $\mathbb{A}^1$ -invariant) motivic complex of Elmanto–Morrow and Bouis; $\mathbf{Z}(i)_{\mathrm{cdh}}$ denotes cdh-motivic cohomology; and $\mathbf{Z}(i)_{\mathbb{A}^1}$ denotes the $\mathbb{A}^1$ -invariant motivic complex of Bachmann–Elmanto–Morrow. These theories are not identified without a stated comparison theorem.

12.1 Consequences of F -smoothness

Corollary 12.1 (Stability under smooth extensions). *Let V be a valuation ring and fix a prime p . Polynomial and étale V -algebras are F -smooth at p , as are smooth V -algebras satisfying the standard p -quasisyntomic hypotheses. The same is true for filtered colimits of such F -smooth rings.*

Proof. Theorem 1.2 supplies F -smoothness of V . Apply [4, Propositions 4.6–4.9], which establish the stated stability under filtered colimits, étale maps, polynomial extensions, and p -completely flat extensions smooth modulo p . $\square$

Theorem 1.2 also allows one to apply the Beilinson t -structure description of prismatic Frobenius in [4, Remark 4.11] to any valuation ring. In particular the Nygaard-filtered Frobenius map is the connective-cover comparison of complete filtered objects in the sense used there. This is a reformulation of the F -smoothness criterion, not an additional completeness argument.

12.2 Cdh-motivic cohomology on arbitrary schemes

Theorem 12.2 (Universal cdh-local F -smoothness). *Every commutative ring is cdh-locally F -smooth in the sense of [20, Definition 6.4].*

Proof. The defining condition is that every valuation ring over the given ring be F -smooth at all primes. This is Theorem 1.2. $\square$

Corollary 12.3 (Cdh comparison and projective bundles). *For any quasi-compact quasi-separated scheme X and $i \geq 0$, the canonical comparison is an equivalence*

$$\mathbf{Z}(i)_{\text{cdh}}(X) \xrightarrow{\sim} \mathbf{Z}(i)_{\mathbb{A}^1}(X).$$

The cdh-motivic complexes are therefore $\mathbb{A}^1$ -invariant on all such schemes and satisfy the projective bundle formula. More precisely, for a rank- d vector bundle $\mathcal{E}$ on X , the usual first-Chern-class map gives

$$\bigoplus_{r=0}^{d-1} \mathbf{Z}(i-r)_{\text{cdh}}(X)[-2r] \xrightarrow{\sim} \mathbf{Z}(i)_{\text{cdh}}(\mathbf{P}(\mathcal{E})),$$

with the customary convention for negative weights.

Proof. Apply [20, Theorem 6.7] to the base ring $\mathbf{Z}$, which is cdh-locally F -smooth by Theorem 12.2. The identification with the $\mathbb{A}^1$ -invariant theory and its projective bundle formula are established there using [1, Theorems 1.10 and 8.20]. $\square$

12.3 A global comparison bound

Theorem 12.4 (Global mixed-characteristic comparison). *For every quasi-compact quasi-separated scheme X , every prime p , and every $i \geq 0$, the cofiber of the canonical comparison*

$$\mathbf{Z}(i)_{\mathbb{A}^1}(X)/p \longrightarrow R\Gamma_{\text{ét}}(X[1/p], \mu_p^{\otimes i})$$

belongs to $D^{\geq i-1}(\mathbf{F}_p)$. By Corollary 12.3, the same holds with $\mathbf{Z}(i)_{\text{cdh}}$ in place of $\mathbf{Z}(i)_{\mathbb{A}^1}$.

Proof. Theorem 1.1 verifies the hypothesis $\text{Val}(\text{Spec } \mathbf{Z}, p, j)$ for every j . Apply [1, Theorem 1.13] with base $S = \text{Spec } \mathbf{Z}$, which has finite valuative dimension. The cdh replacement follows from Corollary 12.3. $\square$

Remark 12.5. The bound here is $i - 1$ and the coefficients are modulo p . The assertion concerns $\mathbb{A}^1$ -invariant (equivalently, cdh-) motivic cohomology. It does not assert a Beilinson–Lichtenbaum comparison for $\mathbf{Z}(i)_{\text{mot}}$ over arbitrary regular rings.

12.4 Motivic homotopy invariance over Prüfer domains

Theorem 12.6 (Integral motivic homotopy invariance). *Let B be a Prüfer domain, let X be an ind-smooth B -scheme, and let $m \geq 1$. For every $i \geq 0$ the natural map*

$$\mathbf{Z}(i)_{\text{mot}}(X) \xrightarrow{\sim} \mathbf{Z}(i)_{\text{mot}}(\mathbb{A}_X^m)$$

is an equivalence in $D(\mathbf{Z})$. On this class of schemes the canonical maps also induce equivalences

$$\mathbf{Z}(i)_{\text{mot}}(X) \simeq \mathbf{Z}(i)_{\text{cdh}}(X) \simeq \mathbf{Z}(i)_{\mathbb{A}^1}(X).$$

Proof. Theorem 12.2 makes B cdh-locally F -smooth. Apply [20, Theorem 6.9 and its proof]; the last equivalence is Corollary 12.3. $\square$

12.5 Finite-coefficient motivic cohomology over valuation rings

Theorem 12.7 (Generic-fiber calculation). *Let V be a p -torsion-free valuation ring and R a henselian local V -algebra that is ind-smooth over V . For integers $i, n \geq 0$ and $r \geq 1$ there are natural identifications*

$$H^n(R, \mathbf{Z}/p^r(i)_{\text{mot}}) \simeq \begin{cases} H_{\text{ét}}^n(R[1/p], \mu_{p^r}^{\otimes i}), & n < i, \\ \text{im}\left((R^\times)^{\otimes i} \rightarrow H_{\text{ét}}^i(R[1/p], \mu_{p^r}^{\otimes i})\right), & n = i, \\ 0, & n > i. \end{cases}$$

Here the arrow in the middle line is the étale symbol map.

Proof. Theorem 1.2 makes V F -smooth at p . The statement is then exactly [20, Theorem C], which uses [4, Theorem 1.8] and the Beilinson–Lichtenbaum comparison [20, Theorem 5.1] for ind-smooth algebras over valuation rings. $\square$

Corollary 12.8 (Improved Milnor K -theory). *Let V be a henselian p -torsion-free valuation ring, and let $i \geq 0$ and $r \geq 1$. The Galois symbol gives an injection*

$$\widehat{K}_i^M(V)/p^r \hookrightarrow H_{\text{ét}}^i(V[1/p], \mu_{p^r}^{\otimes i}),$$

whose image is the subgroup generated by symbols of units of V . If p is a nonzero nonunit and V has rank one, then $V[1/p] = \text{Frac}(V)$.

Proof. Apply Theorem 12.7 with $R = V$ and $n = i$. By [22, Theorem 2.21], the improved Milnor group modulo p^r identifies with $H^i(V, \mathbf{Z}(i)_{\text{mot}})/p^r$. Furthermore, [21, Theorem 5.1] proves that the integral motivic complex of a henselian valuation ring is concentrated in degrees at most i . The coefficient sequence therefore identifies this quotient with $H^i(V, \mathbf{Z}/p^r(i)_{\text{mot}})$, so the corollary follows. $\square$

Remark 12.9 (Syntomic nearby cycles). For a p -torsion-free F -smooth scheme X and $j : X[1/p] \hookrightarrow X$, [4, Theorem 1.8] identifies $\mathbf{Z}/p^r(i)_{\text{syn}, X} \rightarrow \tau_{\leq i} Rj_* \mu_{p^r}^{\otimes i}$ in cohomological degrees below i , and makes the map on H^i injective with image generated by unit symbols. Corollary 12.1 supplies this for smooth schemes over a p -torsion-free valuation base when the hypotheses of that theorem hold.

These applications concern different coefficient conventions and categories of schemes. They do not, without further arguments, prove the integral Gersten conjecture, a regular-base Beilinson–Lichtenbaum equivalence for the non- $\mathbb{A}^1$ -invariant theory, or additional pro-cdh comparison statements.

References

- [1] T. Bachmann, E. Elmanto, and M. Morrow, $\mathbb{A}^1$ -invariant motivic cohomology of schemes, [arXiv:2508.09915](#).
- [2] B. Bhatt and J. Lurie, *Absolute prismatic cohomology*, [arXiv:2201.06120](#).
- [3] B. Bhatt and J. Lurie, *The prismatization of p -adic formal schemes*, [arXiv:2201.06124](#).
- [4] B. Bhatt and A. Mathew, *Syntomic complexes and p -adic étale Tate twists*, [arXiv:2202.04818](#).
- [5] B. Bhatt, M. Morrow, and P. Scholze, *Integral p -adic Hodge theory*, Publ. Math. Inst. Hautes Études Sci. **128** (2018), 219–397; [arXiv:1602.03148](#).
- [6] B. Bhatt, M. Morrow, and P. Scholze, *Topological Hochschild homology and integral p -adic Hodge theory*, Publ. Math. Inst. Hautes Études Sci. **129** (2019), 199–310; [arXiv:1802.03261](#).
- [7] B. Bhatt and P. Scholze, *Prisms and prismatic cohomology*, Ann. of Math. **196** (2022), 1135–1275; [arXiv:1905.08229](#).
- [8] O. Gabber and L. Ramero, *Almost Ring Theory*, Lecture Notes in Mathematics 1800, Springer, 2003; [arXiv:math/0201175](#).
- [9] T. Geisser and M. Levine, *The K -theory of fields in characteristic p* , Invent. Math. **139** (2000), 459–493.
- [10] A. Hattori, *A foundation of torsion theory for modules over general rings*, Nagoya Math. J. **17** (1960), 147–158.
- [11] M. Kerz, *Milnor K -theory of local rings with finite residue fields*, J. Algebraic Geom. **19** (2010), 173–191.
- [12] M. Lüders and M. Morrow, *Milnor K -theory of p -adic rings*, [arXiv:2101.01092](#).
- [13] Z. Mao, *Frobenius–Witt cotangent complex for derived rings*, [arXiv:2608.04487v3](#).
- [14] M. Morrow, *Pro unitality and pro excision in algebraic K -theory and cyclic homology*, J. Reine Angew. Math. **736** (2018), 95–139; [arXiv:1404.4179](#).
- [15] G. Tamme, *Excision in algebraic K -theory revisited*, Compos. Math. **154** (2018), 1801–1814; [arXiv:1703.03331](#).
- [16] L. Tang, *Syntomic cycle classes and prismatic Poincaré duality*, [arXiv:2210.14279](#).
- [17] W. van der Kallen, *The K_2 of rings with many units*, Ann. Sci. École Norm. Sup. (4) **10** (1977), 473–515.
- [18] V. Voevodsky, *On motivic cohomology with $\mathbf{Z}/\ell$ -coefficients*, Ann. of Math. **174** (2011), 401–438; [arXiv:0805.4430](#).
- [19] S. Kelly and M. Morrow, *K -theory of valuation rings*, Compos. Math. **157** (2021), 1121–1142; [arXiv:1810.12203](#).
- [20] T. Bouis and A. Kundu, *Beilinson–Lichtenbaum phenomenon for motivic cohomology*, [arXiv:2506.09910](#); see also the author’s [revised manuscript](#).
- [21] T. Bouis, *On the motivic cohomology of some singular rings*, [preprint](#) (2026).
- [22] T. Bouis, *Weibel vanishing and the projective bundle formula for mixed characteristic motivic cohomology*, [arXiv:2507.16501](#).